Bayesian Quantile Regression with Deep Echo State Network Features

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Abstract

Conditional quantiles are central to asymmetric decision losses, tail-risk assessment, and interval forecasts, but Bayesian quantile regression for nonlinear dynamic series can be difficult when the temporal feature vector is high-dimensional. We develop the quantile deep echo state network (Q–DESN), a Bayesian quantile regression model conditional on fixed features generated by a deep echo state network. After the reservoir architecture, random seed, scaling, reducers, and washout convention are fixed, posterior uncertainty is assigned to the regression coefficients, likelihood, and shrinkage parameters. Single-level fits use asymmetric Laplace or quantile-fixed generalized asymmetric Laplace working likelihoods with ridge or regularized-horseshoe priors. Posterior computation uses Markov chain Monte Carlo where feasible and a model-specific variational Bayes approximation for lower-cost screening, initialization, and selected applications. For quantile grids, we compare independent level-wise regressions followed by monotone rearrangement with a joint quantile-vector regression that shrinks adjacent quantile-specific coefficient differences. Synthetic studies, a single-origin retrospective GloFAS streamflow case, and a retrospective PriceFM comparison identify settings where this fixed-feature Bayesian regression improves finite-grid quantile scores. These findings are design-specific: the displayed bands are model-based quantile summaries, the score denoted aCRPS is a finite-grid integral of check loss over the fitted quantile levels, and calibration, quantile crossing, and reservoir sensitivity remain empirical diagnostics.

1 Introduction

Forecast evaluation often requires summaries beyond the conditional mean. Conditional means are natural under squared-error loss, but asymmetric decision losses, tail-risk assessment, and interval forecasts call for conditional quantiles. Under asymmetric piecewise-linear loss, conditional quantiles are optimal point forecasts, and pairs of quantiles define interval forecasts (Koenker and Bassett, 1978; Koenker, 2005; Gneiting, 2011). Let Y_{t+h} denote the future response at horizon h, and let $\mathcal{F}_t$ be the sigma-field generated by the information available at forecast origin t, including past responses and covariates known at the origin or specified through a scenario. With $F_{t,h}(u) = \Pr(Y_{t+h} \le u \mid \mathcal{F}_t)$, define the level-$p_0$ conditional quantile by the generalized inverse

$$Q_{t,h}(p_0) = F_{t,h}^{-1}(p_0) = \inf\{u : F_{t,h}(u) \ge p_0\}, \qquad p_0 \in (0,1).$$

When the forecasting question concerns one probability level, a single fitted conditional quantile is the relevant summary. When several levels are fitted, the resulting finite grid can be rearranged into a monotone quantile curve. Such a curve is a predictive summary over the chosen grid. A full predictive distribution additionally requires a construction that couples the levels into a distribution (Gneiting and Raftery, 2007; Gneiting and Katzfuss, 2014; Barlow and Brunk, 1972; Chernozhukov et al., 2010).

For nonlinear dynamic data, quantile forecast accuracy also depends on how temporal dependence is represented. An echo state network (ESN) represents past responses and covariates through a recurrent nonlinear state equation. In the construction used here, the input and recurrent weights are generated once and then held fixed. Running this fixed recurrence through the observed history produces reservoir states, which are then used as rows of a nonlinear design matrix for regression. The ESN literature calls the final regression layer the readout (Jaeger, 2001; Lukoševičius and Jaeger, 2009; Lukoševičius, 2012). A deep echo state network (DESN) extends this idea by stacking reservoir layers, following the deep ESN construction of Gallicchio et al. (2018). Conditional on the selected architecture, random seed, scaling constants, reducers, and washout convention, the DESN therefore supplies a deterministic nonlinear feature matrix. Q–DESN uses Bayesian quantile regression with that fixed design, conditioning on the fixed recurrent weights throughout the analysis.

The quantile deep echo state network (Q–DESN) combines this fixed DESN design with a quantile working likelihood. For a chosen level p_0, the linear predictor $\mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}$ is the likelihood location and, under the parameterization used here, the fitted p_0 conditional quantile. The baseline likelihood is asymmetric Laplace (AL). The extended asymmetric Laplace likelihood, denoted exAL, is the quantile-fixed generalized asymmetric Laplace construction of Yan et al. (2025); it allows mode, skewness, and tail behavior to vary while retaining the fitted quantile level. In model labels, the prefix "ex" indicates this extended working likelihood, while unprefixed QDESN and DQLM labels use AL.

The AL or exAL likelihood is used as a working likelihood in the Bayesian quantile-regression sense. It defines a posterior distribution for quantile-specific coefficients without asserting that either likelihood is the data-generating error law (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). Posterior intervals for coefficient-derived quantiles and forecast bands computed from the fitted likelihood are therefore model-based summaries conditional on the fixed DESN design and the chosen working likelihood. Frequentist coverage of credible intervals under likelihood misspecification is a separate question from empirical coverage of the forecast bands; the simulations and retrospective applications assess the latter empirically.

High-dimensional regressions on DESN features require regularization. The default coefficient prior is ridge, which gives dense Gaussian shrinkage. As an adaptive alternative, the regularized horseshoe (RHS) introduces global-local shrinkage and a slab component for large coefficients (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, the local, global, and slab scale updates retain closed-form inverse-gamma forms (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). Posterior computation uses Markov chain Monte Carlo (MCMC) as the reference method when the run is feasible. A mean-field variational Bayes (VB) approximation is used for screening, initialization, and selected applications. For non-conjugate exAL scale-asymmetry blocks, the VB implementation uses the Laplace–Delta approximation of Wang and Blei (2013); the label VB–LD refers to this model-specific approximation.

This article makes three principal contributions. First, it introduces Q–DESN as Bayesian conditional-quantile regression on a fixed DESN feature map, with AL or exAL working likelihoods and ridge or regularized-horseshoe regularization. Second, it develops two distinct strategies for quantile grids: independent level-wise regressions followed by a pre-specified monotone rearrangement, and a joint quantile-vector regression that shares information through regularized-horseshoe shrinkage on adjacent coefficient differences, borrowing from Quantile-Varying Parameter ideas (Kohns and Szendrei, 2025). This adjacent-difference prior encourages similar slopes at neighboring quantile levels; exact noncrossing still requires an explicit constraint or monotone rearrangement. Third, the article gives MCMC and model-specific VB–LD computation and evaluates the resulting procedures in simulations and two retrospective applications. The applications are interpreted under their recorded covariates and forecast origins.

Construction	Estimated quantity	Fitting structure	Forecast summary
Single-level Q–DESN	$Q_{p0}(Y_t \mid \mathcal{F}_t)$	AL or exAL working likelihood with ridge or RHS coefficient shrinkage; MCMC or VB	One fitted conditional quantile; any response simulation is tied to the stated forecast design.
Independent-grid Q–DESN	$\{Q_{p_k}(Y_t \mid \mathcal{F}_t)\}_{k=1}^{K}$	Separate single-level fits at each quantile level, followed by a pre-specified monotone rearrangement	Quantile grid and interpolated monotone curve with level-wise posterior summaries.
Joint Q–DESN	Quantile vector over a fixed grid	Composite AL or exAL working likelihood with RHS shrinkage on baseline slopes and adjacent quantile-indexed slope increments; MCMC is the reference computation	Jointly estimated raw quantile path; exact monotonicity still requires a hard constraint or monotone rearrangement.

Table 1: Model map. AL versus exAL is a working-likelihood choice, ridge versus RHS is a coefficient-prior choice, independent versus joint fitting is a multi-quantile strategy, and MCMC versus variational Bayes is a computational choice.

What the empirical studies evaluate. The empirical studies address distinct questions. The single-quantile simulation compares QDESN and exQDESN regressions with DQLM and exDQLM baselines. The joint simulation evaluates multi-quantile coefficient sharing against independent level-wise regressions under known true conditional quantile paths. The GloFAS and PriceFM examples are retrospective applications of selected independent Q–DESN regressions. Evidence for the joint quantile-vector prior comes from the joint simulation study.

Related Work

Bayesian quantile regression provides the likelihood-based foundation for Q–DESN. Classical quantile regression targets conditional quantiles directly (Koenker and Bassett, 1978; Koenker, 2005). Bayesian implementations often use the asymmetric Laplace likelihood because its mode aligns with check-loss minimization and because its mixture representation supports conditional posterior computation (Yu and Moyeed, 2001; Kozumi and Kobayashi, 2011). In this article the asymmetric likelihood is a working likelihood: it defines posterior uncertainty for quantile-regression parameters, while frequentist coverage of model-based credible intervals under misspecification may require separate assessment or adjustment under misspecification (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) relaxes the AL error-shape restriction after the fitted quantile level has been fixed; this is the exAL working likelihood used below.

Dynamic quantile linear models provide Bayesian state-space baselines for time-varying conditional quantiles. The dynamic quantile linear model (DQLM) combines a linear state evolution with Bayesian quantile regression (Gonçalves et al., 2020). The extended DQLM (exDQLM) of Barata et al. (2022) adds the corresponding asymmetric-likelihood flexibility, and both paths are implemented in the CRAN package exdqlm (Barata et al., 2026). Q–DESN changes the representation of temporal dependence: rather than specifying a low-dimensional linear state process, it uses a fixed nonlinear reservoir state as the regression design. The DQLM and exDQLM fits therefore serve as structured linear dynamic quantile comparators under the same held-out forecast evaluation.

Reservoir computing and ESNs provide the nonlinear dynamic features used by Q–DESN. ESNs were introduced by Jaeger (2001) and later reviewed within the broader reservoir-computing literature by Lukoševičius and Jaeger (2009); practical guidance is given by Lukoševičius (2012). In a leaky-integrator ESN, the leak rate controls how strongly the current state moves toward the new input-driven update rather than persisting from the previous state. Analyses of the echo state property caution that spectral-radius heuristics should be checked rather than treated as sufficient stability guarantees (Jaeger et al., 2007; Yildiz et al., 2012). Broader reservoir-computing surveys and time-series reviews discuss their role in forecasting and applied dynamic modeling (Zhang and Vargas, 2023; Yan et al., 2024; Cardoso et al., 2024; Sun et al., 2024). Deep ESNs, topology variants, and reservoir hyperparameter tuning extend the basic construction by stacking reservoirs, changing connectivity, or selecting reservoir parameters (Gallicchio et al., 2018; Kim and King, 2020; Maat et al., 2018; Bai et al., 2023; Viehweg et al., 2025). Q–DESN uses these reservoir constructions as fixed nonlinear features for Bayesian quantile regression.

A related reservoir-computing literature embeds ESN features in probabilistic forecasting models. Deep ESNs with uncertainty quantification, probabilistic reservoir methods for load forecasting, activation-level uncertainty propagation, and Bayesian or state-space ESN formulations show that reservoir features can support uncertainty-aware prediction (McDermott and Wikle, 2019; Guerra et al., 2023; Gandhi et al., 2018; Singh and Raman, 2025). The narrower gap addressed here is Bayesian inference for conditional quantiles with fixed DESN features, asymmetric working likelihoods, and high-dimensional coefficient regularization.

Quantile-oriented ESN methods provide the closest forecasting comparison. Lv et al. (2016) use quantile-regression ESN ensembles to construct prediction intervals for blast-furnace gas generation, and Bonas et al. (2024) use penalized quantile regression to calibrate DESN forecasts for quasi-periodic climate processes. Q–DESN is complementary to these approaches. Conditional on the generated reservoir, it specifies an exAL regression whose posterior includes coefficients, likelihood scale, asymmetry, and shrinkage parameters. The joint extension further couples quantile-specific regression coefficients during inference, so raw multi-quantile paths can be more coherent before monotone rearrangement is applied.

Shrinkage priors enter because fixed reservoir designs can be high-dimensional and correlated. Ridge shrinkage provides the default dense Gaussian regularization. The regularized horseshoe, building on the horseshoe prior and its slab-regularized extension, is used as an adaptive global-local alternative (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, scale updates remain closed form (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). These priors regularize the coefficients. Feature-level interpretability requires additional diagnostics.

Finally, a grid of quantile levels requires a distinction between monotone rearrangement after fitting and joint modeling across levels. Separate Q–DESN fits can cross because each level is estimated independently. Isotonic regression can project the fitted grid onto the monotone cone (Barlow and Brunk, 1972), and rearrangement methods provide a general approach to noncrossing quantile and probability curves (Chernozhukov et al., 2010). These operations repair the displayed quantile curve after fitting. A joint posterior over quantile-indexed coefficients requires a joint model across levels. Constrained quantile-regression methods instead impose non-crossing restrictions during estimation (Wu and Liu, 2009; Bondell et al., 2010). Recent work also shows that non-crossing constraints can be viewed as fused shrinkage across quantile-specific coefficients (Szendrei et al., 2025). Bayesian joint quantile models share information across levels through process priors, score-likelihood constructions, or smoothness penalties (Yang and Tokdar, 2017; Wu and Narisetty, 2021; Wang and Cai, 2024). The Quantile-Varying Parameter construction of Kohns and Szendrei (2025) is especially relevant here because it places a state-space shrinkage prior on adjacent quantile-indexed coefficient changes. The joint Q–DESN extension below adapts that idea to fixed-feature regressions and assigns regularized-horseshoe priors to adjacent slope differences. The resulting prior encourages neighboring quantile curves to have similar slopes, which can reduce crossings. In the validation study below, it substantially reduces raw crossings relative to independent AL regressions before monotone rearrangement; exact monotonicity still requires an explicit ordered parameterization, projection, or rearrangement.

These distinctions determine the notation used below. We first separate the objects fixed by reservoir construction from the stochastic coefficient and likelihood quantities, so that posterior statements can be read as conditional on a recorded feature map.

The remainder of the paper is organized as follows. Section 2 introduces notation, deep echo

Symbol	Description	Dimension and support
y_t	scalar response at time t	scalar
$p_0,\ p_k$	single-level and quantile-grid probabilities	$(0,1)$
$\boldsymbol{z}_t$	exogenous covariates	$q \times 1$
$\boldsymbol{u}_t$	response lags and exogenous covariates	$(m+q) \times 1$
$\boldsymbol{h}_{t,d}$	reservoir state at layer d	$n_d \times 1$
$\tilde{\boldsymbol{h}}_{t,d}$	reduced state at layer d	$\tilde{n}_d \times 1$
α_d, ρ_d	layer leak and target spectral radius	$[0,1]$ and $(0,1)$
$\tilde{\boldsymbol{x}}_t$	feature vector without intercept	$r \times 1$
$\boldsymbol{x}_t$	DESN design vector with intercept	$(r+1) \times 1$
$\boldsymbol{X}$	fixed DESN design matrix after washout	$T \times (r+1)$
$\boldsymbol{\beta}$	quantile-regression coefficients	$(r+1) \times 1$
κ_β^2	default ridge slope-prior variance	positive
σ, γ	exAL scale and asymmetry parameters	$\sigma > 0,\ \gamma \in (L, U)$
(λ_j, τ, ζ)	local, global, and slab shrinkage scales	positive

Table 2: Key notation used throughout. Reservoir quantities are drawn or constructed once, held fixed after washout, and collected in the design matrix $\boldsymbol{X}$.

state network (DESN) dynamics, and the exAL representation. Section 3 presents the Q–DESN model and prior specification, including independent and joint multi-quantile extensions. Section 4 describes posterior computation, while Sections 5–6 describe multi-step quantile summaries, finite-grid scoring, and reservoir diagnostics. Section 7 presents the simulation design, and Sections 8–9 give reproducible streamflow and electricity-price forecasting applications. Section 10 summarizes limitations and future work. The supplement gives full conditional derivations, variational and ELBO details, monotone rearrangement details, and additional validation tables.

2 Notation and Preliminaries

The notation separates deterministic reservoir quantities from stochastic coefficient and likelihood quantities. Table 2 summarizes the recurring objects. Matrices and vectors are bold, as in $\boldsymbol{W}$ and $\boldsymbol{x}$; scalar observations, latent states, and parameters use standard lowercase notation unless they are named constants. The symbols IG, GIG, and $\mathcal{N}^+$ denote inverse-gamma, generalized inverse Gaussian, and positive-truncated Normal distributions. For a square matrix $\boldsymbol{A}$, write $\rho(\boldsymbol{A}) = \max_i |\lambda_i(\boldsymbol{A})|$, where $\{\lambda_i(\boldsymbol{A})\}$ are the eigenvalues. Thus $\rho(\cdot)$ denotes spectral radius, while $\rho_d \in (0,1)$ denotes the target spectral radius for layer d. Indices use $d = 1, \ldots, D$ and $t = 1, \ldots, T$. When conditioning information is shown explicitly, $Q_p(y \mid \cdot)$ denotes the conditional p-quantile of the scalar response. The identity matrix, zero vector, and one vector of dimension m are denoted by $\boldsymbol{I}_m$, $\mathbf{0}_m$, and $\mathbf{1}_m$, respectively.

2.1 Deep Echo State Network (DESN)

The DESN is a deterministic feature map once the architecture, random seed, scaling constants, reducers, and washout convention are fixed. Conditional on these choices, posterior inference is

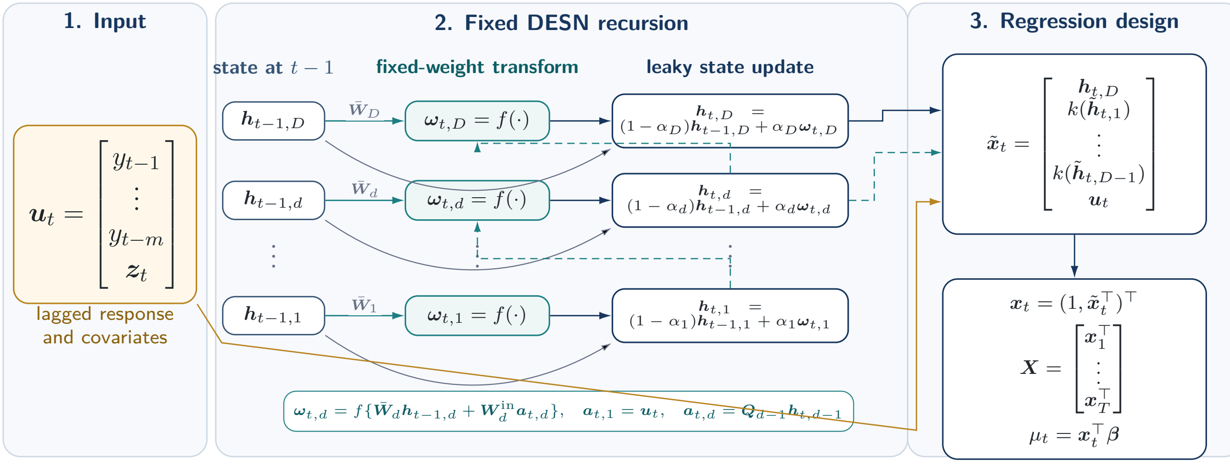

Figure 1: Fixed DESN feature construction. The input block collects response lags and exogenous covariates. The reservoir block shows that each layer update combines the previous state with a nonlinear innovation driven by fixed recurrent weights and, for upper layers, the current-time reduced state from the layer below. After washout, the top-layer state, transformed lower-layer reductions, and retained input block form the fixed regression design used by the Q–DESN working likelihood.

restricted to the quantile regression model. The following notation states the dimensions before the recursion so that each layer-specific matrix has a defined statistical role.

Notation and dimensions. The input vector is $\boldsymbol{u}_t = (y_{t-1}, \ldots, y_{t-m}, \boldsymbol{z}_t^\top)^\top \in \mathbb{R}^{m+q}$, with m response lags and q exogenous variables. For layer d, $\boldsymbol{W}_d \in \mathbb{R}^{n_d \times n_d}$ is recurrent, $\boldsymbol{W}_1^{\text{in}} \in \mathbb{R}^{n_1 \times (m+q)}$ maps the original input to the first layer, and $\boldsymbol{W}_d^{\text{in}} \in \mathbb{R}^{n_d \times \tilde{n}_{d-1}}$ maps the reduced state from layer $d-1$ to layer d for $d \geq 2$. The reducer $\boldsymbol{Q}_{d-1} \in \mathbb{R}^{\tilde{n}_{d-1} \times n_{d-1}}$ maps $\boldsymbol{h}_{t,d-1} \in \mathbb{R}^{n_{d-1}}$ to $\tilde{\boldsymbol{h}}_{t,d-1} \in \mathbb{R}^{\tilde{n}_{d-1}}$. The regression dimension is $r = n_D + \sum_{d=1}^{D-1} \tilde{n}_d + (m+q)$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_r)^\top \in \mathbb{R}^{r+1}$.

Random sparse weights and spectral scaling. In contrast to fully trained recurrent networks, reservoir and input weights are drawn once and kept fixed throughout inference. Let $\circ$ denote the Hadamard product. Each recurrent matrix is $\boldsymbol{W}_d = \Phi_d^{(W)} \circ Z_d^{(W)}$, and each input matrix is generated analogously as $\boldsymbol{W}_d^{\text{in}} = \Phi_d^{(\text{in})} \circ Z_d^{(\text{in})}$. Mask entries are Bernoulli with probabilities $\pi_d^{(W)}$ and $\pi_d^{(\text{in})}$; base entries are independent draws from zero-mean finite-variance distributions p_W and p_{in}, such as $\mathcal{N}(0, \sigma^2)$ or Unif$[-a, a]$. Masks and base draws are mutually independent and independent across entries and layers. The recurrent matrices are then scaled to the target spectral radii,

$$\text{Spectral-radius scaling:} \qquad \bar{\boldsymbol{W}}_d \;=\; \frac{\rho_d}{\rho(\boldsymbol{W}_d)}\,\boldsymbol{W}_d, \qquad \rho_d \in (0,1).$$

If $\rho(\boldsymbol{W}_d) = 0$, the matrix is resampled. Input matrices $\boldsymbol{W}_d^{\text{in}}$ are not spectrally scaled.

DESN recursion and regression features. Figure 1 summarizes the three objects carried by the fixed feature construction: the lagged input, the layerwise reservoir update, and the fixed regression design.

With layer-specific leak rates $\alpha_d \in [0,1]$, elementwise nonlinearity $f(\cdot)$, and lower-layer transformation $k(\cdot)$, the fixed feature map is

$$\begin{aligned}
\text{Layer 1 innovation:}\quad & \boldsymbol{\omega}_{t,1} = f\Big(\bar{\boldsymbol{W}}_1 \boldsymbol{h}_{t-1,1} + \boldsymbol{W}_1^{\text{in}} \boldsymbol{u}_t\Big), \\
\text{Layer 1 update:}\quad & \boldsymbol{h}_{t,1} = (1-\alpha_1)\boldsymbol{h}_{t-1,1} + \alpha_1 \boldsymbol{\omega}_{t,1}, \\
\text{Reduced bridge state:}\quad & \tilde{\boldsymbol{h}}_{t,d-1} = \boldsymbol{Q}_{d-1} \boldsymbol{h}_{t,d-1}, && d = 2,\ldots,D, \\
\text{Layer d innovation:}\quad & \boldsymbol{\omega}_{t,d} = f\Big(\bar{\boldsymbol{W}}_d \boldsymbol{h}_{t-1,d} + \boldsymbol{W}_d^{\text{in}} \tilde{\boldsymbol{h}}_{t,d-1}\Big), && d = 2,\ldots,D, \\
\text{Layer d update:}\quad & \boldsymbol{h}_{t,d} = (1-\alpha_d)\boldsymbol{h}_{t-1,d} + \alpha_d \boldsymbol{\omega}_{t,d}, && d = 2,\ldots,D, \\
\text{Feature stack:}\quad & \tilde{\boldsymbol{x}}_t = \big[\boldsymbol{h}_{t,D}^\top,\ k(\tilde{\boldsymbol{h}}_{t,1})^\top, \ldots, k(\tilde{\boldsymbol{h}}_{t,D-1})^\top,\ \boldsymbol{u}_t^\top\big]^\top, \\
\text{Intercept and linear predictor:}\quad & \boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top, \qquad \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}.
\end{aligned} \tag{2.1}$$

Here $k(\cdot)$ is an elementwise activation applied to reduced lower-layer states, commonly $k = \text{id}$ or $k = f$.

The input vector $\boldsymbol{u}_t$ bundles response lags and exogenous covariates; the intercept is appended only in $\boldsymbol{x}_t$. The leaky update is a convex combination of the previous state and the current nonlinear update: small α_d retains more memory, while large α_d reacts more quickly to the current input. The leak rate α_d and target radius ρ_d therefore control the empirical time scale and persistence of the deterministic state recursion; they are tuning choices rather than sufficient guarantees of the echo state property. Reducers $\boldsymbol{Q}_{d-1}$ may be fixed random projections, optionally orthonormalized, or precomputed linear reducers such as PCA from an initial preprocessing pass.

The feature stack concatenates the top-layer state, transformed reduced lower-layer states, and $\boldsymbol{u}_t$, yielding $\tilde{\boldsymbol{x}}_t \in \mathbb{R}^r$. Appending an intercept gives $\boldsymbol{x}_t \in \mathbb{R}^{r+1}$, and the coefficient vector maps this feature vector to the scalar location μ_t. States are initialized at $\boldsymbol{h}_{0,d} = \mathbf{0}_{n_d}$, or from $\mathcal{N}(\mathbf{0}_{n_d}, \sigma^2_{h,d}\boldsymbol{I}_{n_d})$. A teacher-forcing washout $t = 1,\ldots,T_0$ is run before features enter the regression; this discards early transient states that depend strongly on the arbitrary initialization. Washout length and scaling choices can affect short-horizon behavior. All masks and base matrices are fixed for the entire analysis and are not resampled during posterior inference.

2.2 Extended Asymmetric Laplace Working Likelihood

The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) extends the asymmetric Laplace (AL) working likelihood while keeping the target quantile fixed. We denote this likelihood by exAL, for extended asymmetric Laplace. The standard AL likelihood is recovered as a special case, but its skewness is fixed once p_0 is chosen. The exAL asymmetry parameter allows mode, skewness, and tail behavior to vary within a quantile-fixed likelihood. At fixed level p_0, μ is the p_0th quantile, $\sigma > 0$ is a scale parameter, and γ is an asymmetry parameter with bounded support (L, U). The endpoints L and U solve $g(\gamma) = 1 - p_0$ and $g(\gamma) = p_0$, respectively, where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and Φ is the standard Normal CDF.

Posterior computation uses the stochastic representation of the exAL distribution. If

$$y \sim \mathrm{exAL}_{p_0}(\mu, \sigma, \gamma),$$

then there exist independent $\epsilon \sim \mathcal{N}(0,1)$, $z \sim \mathrm{Exp}(1)$, and $s \sim \mathcal{N}^+(0,1)$ such that

$$y = \mu + \lambda(\gamma)\sigma s + A(\gamma)\sigma z + \{B(\gamma)\sigma^2 z\}^{1/2}\epsilon,$$

where $A(\gamma) = \{1 - 2p_\gamma\}/\{p_\gamma(1-p_\gamma)\}$, $B(\gamma) = 2/\{p_\gamma(1-p_\gamma)\}$, $C(\gamma) = \{\mathbf{1}\{\gamma > 0\} - p_\gamma\}^{-1}$, $p_\gamma = \mathbf{1}\{\gamma < 0\} + \{p_0 - \mathbf{1}\{\gamma < 0\}\}/g(\gamma)$, and $\lambda(\gamma) = C(\gamma)|\gamma|$. Setting $v = \sigma z$ gives $v \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma)$ and the conditional Gaussian form used for Q–DESN posterior computation:

$$y \mid \mu, \sigma, \gamma, s, v \sim \mathcal{N}(\mu + \lambda(\gamma)\sigma s + A(\gamma)v,\ B(\gamma)\sigma v).$$

All quantities $g(\gamma)$, p_γ, $A(\gamma)$, $B(\gamma)$, $C(\gamma)$, and $\lambda(\gamma)$ depend on the target quantile level p_0; this dependence is suppressed for readability. Once (p_0, γ) is fixed, these are deterministic constants in the Gaussian augmentation.

3 Model Specification

Fix a quantile level $p_0 \in (0,1)$. After reservoir construction and washout, $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T)^\top$ is treated as fixed, and all posterior uncertainty is conditional on this design. The statistical model is a Bayesian quantile regression for $\boldsymbol{y}$ given $\boldsymbol{X}$. The observation model and priors define the posterior distribution; the Gaussian augmentation below is a computational representation of the asymmetric working likelihood. Dependence on p_0 is suppressed when no ambiguity arises.

3.1 Quantile Deep Echo State Network (Q–DESN)

The Q–DESN is obtained by inserting the fixed DESN feature map into the exAL working likelihood of Section 2.2. Let $\tilde{\boldsymbol{x}}_t$ be the feature vector in Eq. (2.1), and set $\boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top$. Conditional on the fixed design matrix, the marginal working likelihood is

$$y_t \mid \boldsymbol{x}_t, \boldsymbol{\beta}, \sigma, \gamma \sim \mathrm{exAL}_{p_0}(\boldsymbol{x}_t^\top \boldsymbol{\beta}, \sigma, \gamma), \qquad t = 1, \ldots, T.$$

Thus the dynamic linear predictor $\boldsymbol{x}_t^\top \boldsymbol{\beta}$ is the model's conditional p_0-quantile, while σ and γ control the scale and shape of the working error distribution. For posterior computation, the same exAL mixture representation is applied at each time point:

$$
\begin{aligned}
\text{Quantile linear predictor :} \quad & \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}, \qquad \boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_r)^\top \in \mathbb{R}^{r+1}, \\
\text{Gaussian conditional :} \quad & y_t \mid \boldsymbol{\beta}, \sigma, \gamma, s_t, v_t, \boldsymbol{x}_t \sim \mathcal{N}\Big(\mu_t + \lambda(\gamma)\sigma s_t + A(\gamma) v_t,\, B(\gamma)\sigma v_t\Big), \\
\text{Positive-shift latent :} \quad & s_t \sim \mathcal{N}^+(0,1), \\
\text{Scale-mixture latent :} \quad & v_t \mid \sigma \sim \mathrm{Exp}(\text{rate} = 1/\sigma).
\end{aligned}
\tag{3.1}
$$

The display separates the statistical quantile model from the computational augmentation. The first line defines the quantile path; the second gives the conditional Gaussian observation equation; and the last two lines introduce auxiliary variables. The constants $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$ are inherited from Section 2.2. The DESN feature map is fixed before posterior inference, while $\boldsymbol{\beta}$, $\sigma > 0$, and $\gamma \in (L, U)$ are posterior unknowns. Integrating out (s_t, v_t) recovers the marginal exAL likelihood. These variables expose conditionally Gaussian updates while preserving the quantile target. Complete-data joint distributions for the AL and exAL likelihoods with ridge and RHS priors are given in Section S4 of the supplement.

3.2 Single-Level Prior Specification

The intercept is modeled separately from the slope coefficients: $\beta_0 \sim \mathcal{N}(b_0, V_0)$, with $b_0 \in \mathbb{R}$ and $V_0 > 0$. This lets shrinkage act on the high-dimensional DESN features without forcing the overall fitted quantile level toward zero.

Ridge prior. For the non-intercept coefficients $\boldsymbol{\beta}_{1:r} = (\beta_1, \ldots, \beta_r)^\top$, the default ridge prior is

$$
\boldsymbol{\beta}_{1:r} \mid \kappa_\beta^2 \sim \mathcal{N}(\mathbf{0}_r,\, \kappa_\beta^2 \boldsymbol{I}_r), \tag{3.2}
$$

where $\kappa_\beta^2 > 0$ is fixed or assigned a weakly informative scale prior. This Gaussian prior gives dense baseline regularization.

Adaptive shrinkage. The regularized horseshoe (RHS) prior is the adaptive shrinkage alternative. It starts from the ordinary horseshoe (Carvalho et al., 2010), adds slab regularization (Piironen and Vehtari, 2017), and uses the product representation of Nishimura and Suchard (2023) to retain closed-form scale updates. For non-intercept coefficients $j = 1, \ldots, r$, the ordinary horseshoe is

$$
\beta_j \mid \tau, \lambda_j \sim \mathcal{N}(0, \tau^2 \lambda_j^2), \qquad \lambda_j \sim C^+(0,1), \qquad \tau \sim C^+(0, \tau_0), \tag{3.3}
$$

where τ is the global scale and λ_j is coefficient-specific. This prior concentrates mass near zero while allowing some coefficients to escape strong shrinkage. The regularized version introduces a slab scale $\zeta > 0$ to control the largest prior variances and sets

$$
V_j(\lambda_j, \tau, \zeta) = \left(\zeta^{-2} + \tau^{-2}\lambda_j^{-2}\right)^{-1} = \frac{\zeta^2 \tau^2 \lambda_j^2}{\zeta^2 + \tau^2 \lambda_j^2}, \qquad j = 1, \ldots, r, \tag{3.4}
$$

with conditional prior

$$\beta_j \mid \lambda_j, \tau, \zeta \sim \mathcal{N}(0, V_j), \qquad j = 1, \ldots, r. \tag{3.5}$$

This is the RHS coefficient prior used in the adaptive-shrinkage Q–DESN fits. Equation (3.4) gives the interpretation: if $\tau^2\lambda_j^2 \ll \zeta^2$, then $V_j \approx \tau^2\lambda_j^2$, recovering ordinary horseshoe behavior; if $\tau^2\lambda_j^2 \gg \zeta^2$, then $V_j \approx \zeta^2$, so the slab caps the effective prior variance.

The regularized horseshoe can be written directly as in Piironen and Vehtari (2017). For Gibbs and coordinate-ascent variational updates, however, the direct global-local block includes an extra $\{1+\tau^2\lambda_j^2/\zeta^2\}^{1/2}$ factor, so the ordinary horseshoe updates for (τ, λ_j) are not preserved. We therefore use the equivalent product representation of Nishimura and Suchard (2023), which keeps the same conditional coefficient prior while restoring closed-form scale updates:

$$p(\beta_j, \lambda_j \mid \tau, \zeta) \propto \exp\left(-\frac{\beta_j^2}{2\tau^2\lambda_j^2}\right) \exp\left(-\frac{\beta_j^2}{2\zeta^2}\right) \lambda_j^{-1}(1+\lambda_j^2)^{-1}, \tag{3.6}$$

where proportionality is with respect to (β_j, λ_j). The (τ, ζ)-dependent Gaussian normalizing factors are retained in the full-conditionals derivations so that the inverse-gamma shape parameters are recovered correctly. Thus the RHS option gives adaptive global-local shrinkage while retaining inverse-gamma updates for the local and global scales.

Closed-form scale updates use the inverse-gamma representation of half-Cauchy priors (Makalic and Schmidt, 2016):

$$\lambda_j^2 \mid \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \qquad \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, 1\right), \quad j = 1, \ldots, r, \tag{3.7}$$

$$\tau^2 \mid \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \qquad \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\tau_0^2}\right). \tag{3.8}$$

The slab scale is either fixed or assigned

$$\zeta^2 \sim \mathrm{IG}(a_\zeta, b_\zeta). \tag{3.9}$$

Under (3.6), this prior remains compatible with closed-form updating (Nishimura and Suchard, 2023).

The likelihood parameters are assigned $\sigma \sim \mathrm{IG}(a_\sigma, b_\sigma)$ and $\gamma \sim \pi_\gamma(\gamma)\,\mathbf{1}\{L < \gamma < U\}$, with (L, U) determined by the quantile-fixing construction as described in Section 2.2. For RHS fits, collect $\Theta = (\boldsymbol{\beta}, \sigma, \gamma, \boldsymbol{\lambda}^2, \tau^2, \boldsymbol{\nu}, \xi, \zeta^2)$, where $\boldsymbol{\lambda}^2 = (\lambda_1^2, \ldots, \lambda_r^2)^\top$ and $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_r)^\top$. The entry ζ^2 is omitted when the slab scale is fixed. Ridge fits replace the global-local shrinkage block by κ_β^2 when that scale is assigned a prior.

3.3 Joint Quantile-Vector Regression with DESN Features

Joint quantile-regression model. The single-level model can be fit independently over a grid of quantile levels, with separate regression coefficients at each level. When several conditional quantiles are estimated together, let $0 < p_1 < \cdots < p_Q < 1$ be the fixed quantile grid and let $\boldsymbol{Z}$ denote the $T \times r$ non-intercept part of the fixed design matrix. For Q–DESN, $\boldsymbol{Z}$ is obtained from the DESN feature matrix after removing the intercept column. The joint quantile-vector regression separates intercepts from high-dimensional slopes:

$$Q_{p_q}(y_t \mid \mathcal{F}_t) = \alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \qquad q = 1, \ldots, Q, \quad t = 1, \ldots, T, \tag{3.10}$$

where $\alpha_q \in \mathbb{R}$ is the quantile-specific intercept and $\boldsymbol{\beta}_q \in \mathbb{R}^r$ is the slope vector at level p_q. By default, the high-dimensional RHS path is placed on the slope vectors; intercepts are either weakly regularized independently or represented through ordered gaps when a stronger monotonicity convention is desired.

This construction generalizes the single-level RHS Q–DESN by replacing the single intercept-slope coefficient vector $(\beta_0, \boldsymbol{\beta}_{1:r})$ with $\{(\alpha_q, \boldsymbol{\beta}_q)\}_{q=1}^{Q}$. For $Q = 1$, the individual RHS Q–DESN is recovered by taking α_1 as the single-level intercept and identifying $\boldsymbol{\beta}_1$ with the single-level non-intercept slope vector, with no learned slope baseline. Equivalently, set $\boldsymbol{\beta}_{\text{base}} \equiv \mathbf{0}_r$ and place the RHS prior directly on $\Delta_1 = \boldsymbol{\beta}_1$.

Composite working likelihood. For each q, the working likelihood uses the exAL family at the corresponding target level,

$$y_t \mid \alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q \sim \text{exAL}_{p_q}\left(\alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \sigma_q, \gamma_q\right). \tag{3.11}$$

The Q likelihood contributions form a composite working likelihood over the same response series. The joint analyses in this article use this untempered composite posterior. It updates the quantile-indexed coefficient path. Future-response simulation requires the separate forecast construction in Section 5.1.

Shrinkage across adjacent quantile levels. For a multi-quantile grid, the joint prior adapts the Quantile-Varying Parameter idea that slopes at neighboring quantile levels should be similar unless the data support tail-specific differences (Kohns and Szendrei, 2025). Let $\boldsymbol{\beta}_{\text{base}} \in \mathbb{R}^r$ be a baseline slope vector and define adjacent coefficient differences

$$\Delta_{1,j} = \beta_{1,j} - \beta_{\text{base},j}, \qquad \Delta_{q,j} = \beta_{q,j} - \beta_{q-1,j}, \quad q = 2, \ldots, Q.$$

We assign an ordinary RHS prior to the baseline slope and RHS priors with quantile-gap-specific global scales to the adjacent coefficient differences:

$$\beta_{\text{base},j} \mid \lambda_j^0, \tau^0, \zeta_0 \sim \mathcal{N}(0, V_j^0), \qquad V_j^0 = \frac{\zeta_0^2 (\tau^0)^2 (\lambda_j^0)^2}{\zeta_0^2 + (\tau^0)^2 (\lambda_j^0)^2}, \tag{3.12}$$

$$\Delta_{q,j} \mid \lambda_{q,j}^\Delta, \tau_q^\Delta, \zeta_\Delta \sim \mathcal{N}(0, V_{q,j}^\Delta), \qquad V_{q,j}^\Delta = \frac{\zeta_\Delta^2 (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}{\zeta_\Delta^2 + (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}. \tag{3.13}$$

This construction separates globally useful feature effects $\boldsymbol{\beta}_{\text{base}}$ from effects that vary across adjacent quantile levels. It is related to interquantile shrinkage and composite-quantile regularization in frequentist quantile regression (Jiang et al., 2014; Zou and Yuan, 2008), but the shrinkage is placed on the Bayesian quantile-coefficient path conditional on the fixed feature design.

The next display rewrites the Q augmented likelihood contributions as one stacked Gaussian regression conditional on the exAL latent variables and the quantile-specific intercepts. This computational representation is algebraically equivalent to the joint quantile model above. Let $\boldsymbol{\beta}_{\mathrm{st}} = (\boldsymbol{\beta}_1^\top, \ldots, \boldsymbol{\beta}_Q^\top)^\top$, $\boldsymbol{Z}_{\mathrm{st}} = \boldsymbol{I}_Q \otimes \boldsymbol{Z}$, and $\boldsymbol{\alpha}_{\mathrm{st}} = (\alpha_1 \mathbf{1}_T^\top, \ldots, \alpha_Q \mathbf{1}_T^\top)^\top$. Under the exAL augmentation, define

$$d_{q,t} = \lambda(\gamma_q)\sigma_q s_{q,t} + A(\gamma_q) v_{q,t}, \qquad \omega_{q,t} = B(\gamma_q)\sigma_q v_{q,t},$$

and stack these quantities into $\boldsymbol{d}$ and diagonal $\boldsymbol{\Omega} = \mathrm{diag}(\omega_{q,t})$. The conditional Gaussian likelihood is

$$\boldsymbol{y}_{\mathrm{st}} \mid \boldsymbol{\beta}_{\mathrm{st}}, \boldsymbol{\alpha}, \boldsymbol{d}, \boldsymbol{\Omega} \sim \mathcal{N}(\boldsymbol{\alpha}_{\mathrm{st}} + \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} + \boldsymbol{d},\, \boldsymbol{\Omega}), \qquad \boldsymbol{y}_{\mathrm{st}} = \mathbf{1}_Q \otimes \boldsymbol{y}.$$

Let $\boldsymbol{H}$ be the block first-difference matrix such that

$$\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} = (\boldsymbol{\beta}_1^\top, (\boldsymbol{\beta}_2 - \boldsymbol{\beta}_1)^\top, \ldots, (\boldsymbol{\beta}_Q - \boldsymbol{\beta}_{Q-1})^\top)^\top.$$

With $\tilde{\boldsymbol{\beta}}_{\mathrm{base}} = (\boldsymbol{\beta}_{\mathrm{base}}^\top, \mathbf{0}_r^\top, \ldots, \mathbf{0}_r^\top)^\top$ and diagonal $\boldsymbol{D}_\Delta$ containing the RHS adjacent-difference variances $V_{q,j}^{\Delta}$, the conditional prior is proportional to

$$\exp\left[-\frac{1}{2}(\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} - \tilde{\boldsymbol{\beta}}_{\mathrm{base}})^\top \boldsymbol{D}_\Delta^{-1}(\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} - \tilde{\boldsymbol{\beta}}_{\mathrm{base}})\right].$$

Consequently, the Gaussian full conditional for the stacked slope path has precision

$$\boldsymbol{K}_\beta = \boldsymbol{Z}_{\mathrm{st}}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{Z}_{\mathrm{st}} + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \boldsymbol{H}, \tag{3.14}$$

and mean

$$\boldsymbol{m}_\beta = \boldsymbol{K}_\beta^{-1}\left[\boldsymbol{Z}_{\mathrm{st}}^\top \boldsymbol{\Omega}^{-1}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{d}) + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \tilde{\boldsymbol{\beta}}_{\mathrm{base}}\right]. \tag{3.15}$$

The sparse, block-banded prior contribution in (3.14) is the central computational object for the joint extension.

Soft noncrossing. The non-crossing interpretation is local in adjacent quantile gaps. For neighboring levels,

$$Q_{p_q}(y_t \mid \mathcal{F}_t) - Q_{p_{q-1}}(y_t \mid \mathcal{F}_t) = (\alpha_q - \alpha_{q-1}) + \tilde{\boldsymbol{x}}_t^\top(\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}).$$

On a bounded feature domain, for example after an explicit scaling or clipping condition that puts each component of $\tilde{\boldsymbol{x}}_t$ in $[-1, 1]$, the inequality $\alpha_q - \alpha_{q-1} \geq \|\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}\|_1$ is sufficient for non-crossing. The adjacent-difference RHS prior shrinks $\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}$ toward zero, reducing the size of the feature-dependent term and concentrating the posterior near coherent quantile paths. Exact monotonicity still requires an explicit ordered-intercept parameterization, posterior rejection rule, isotonic projection, or monotone rearrangement. The synthetic joint validation below therefore records raw crossings separately from the pre-specified monotone rearrangement used for scored quantile grids.

4 Posterior Inference

4.1 Single-Level Augmented Posterior

Single-level posterior. For one probability level p_0, the Q–DESN posterior conditions on the fixed DESN design and updates only the regression coefficients, working-likelihood, and shrinkage parameters. Let $\boldsymbol{Z}$ denote the non-intercept design, α the intercept, and $\boldsymbol{\beta}_1$ the non-intercept slope vector. Let Θ_Δ collect the local, global, auxiliary, and slab scales for the RHS prior when the adaptive prior is used. For the exAL working likelihood, let $\{v_t, s_t : t = 1, \ldots, T\}$ be the scale-mixture and positive-shift latent variables; for AL, the s_t and γ blocks are omitted. With $\boldsymbol{d}_1$, $\boldsymbol{\Omega}_1$, and ω_t collecting the corresponding complete-data shifts and variances from the augmented likelihood, the single-level complete-data posterior can be written, up to constants not depending on unknowns, as

$$p(\alpha, \boldsymbol{\beta}_1, \sigma, \gamma, \{v_t, s_t\}_{t=1}^{T}, \Theta_\Delta \mid \boldsymbol{y}, \boldsymbol{Z}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{y} - \alpha\mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)^\top \boldsymbol{\Omega}_1^{-1}(\boldsymbol{y} - \alpha\mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)\right]$$
$$\times \prod_{t=1}^{T}\left\{(\omega_t)^{-1/2} p(v_t \mid \sigma) p(s_t)\right\} p(\sigma)\pi_\gamma(\gamma)\mathbf{1}\{L < \gamma < U\} p(\alpha)\, p_\Delta^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_\Delta) p(\Theta_\Delta). \tag{4.1}$$

Here

$$p_\Delta^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_\Delta) \propto \exp\left(-\frac{1}{2}\boldsymbol{\beta}_1^\top \boldsymbol{D}_\Delta^{-1}\boldsymbol{\beta}_1\right)$$

is the ordinary RHS slope prior, with $\Delta_1 = \boldsymbol{\beta}_1$. The ridge case replaces this factor by the Gaussian ridge prior. This display gives the posterior distribution for the individual Q–DESN regression; the joint quantile-vector construction below reuses the same augmented Gaussian structure after stacking levels and replacing the direct slope prior by a regularized-horseshoe prior on quantile-indexed slope increments.

4.2 Joint Quantile-Vector Extension

For a grid $p_1 < \cdots < p_Q$, let Θ_Δ collect the RHS scales for adjacent coefficient differences and let Θ_0 collect the baseline-slope RHS scales. Let Θ_Q collect $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_{\text{st}}$, the likelihood parameters, Θ_Δ, and, for $Q > 1$, $(\boldsymbol{\beta}_{\text{base}}, \Theta_0)$. Let $\boldsymbol{a}_Q = \{v_{q,t}, s_{q,t} : q = 1, \ldots, Q;\ t = 1, \ldots, T\}$ collect the exAL auxiliary variables. With the stacked quantities $\boldsymbol{y}_{\text{st}}$, $\boldsymbol{\alpha}_{\text{st}}$, $\boldsymbol{Z}_{\text{st}}$, $\boldsymbol{d}$, and $\boldsymbol{\Omega}$ defined in Section 3.3, the joint complete-data posterior is

$$
\begin{aligned}
p(\Theta_Q, \boldsymbol{a}_Q \mid \boldsymbol{y}, \boldsymbol{Z}) &\propto \exp\left[-\frac{1}{2}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})^\top \boldsymbol{\Omega}^{-1}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})\right] \\
&\times \prod_{q=1}^{Q}\prod_{t=1}^{T}\left\{(\omega_{q,t})^{-1/2} p(v_{q,t} \mid \sigma_q) p(s_{q,t})\right\} \prod_{q=1}^{Q} p(\sigma_q)\pi_{\gamma_q}(\gamma_q)\mathbf{1}\{L_q < \gamma_q < U_q\} \\
&\times p(\boldsymbol{\alpha})\, p_\Delta^{(Q)}(\boldsymbol{\beta}_{\mathrm{st}} \mid \boldsymbol{\beta}_{\mathrm{base}}, \boldsymbol{D}_\Delta)\, p(\Theta_\Delta)\, \mathcal{B}_Q,
\end{aligned}
\tag{4.2}
$$

where

$$
\mathcal{B}_Q = \begin{cases} p(\boldsymbol{\beta}_{\mathrm{base}} \mid \Theta_0) p(\Theta_0), & Q > 1, \\ 1, & Q = 1. \end{cases}
$$

The Gaussian prior factor $p_\Delta^{(Q)}$ is induced by the first-difference matrix $\boldsymbol{H}$ and difference covariance $\boldsymbol{D}_\Delta$ in (3.14)–(3.15). For $Q > 1$, it is the adjacent-difference prior centered on the baseline slope, so the $\boldsymbol{\beta}_{\mathrm{base}}$ argument is active. The AL special case omits $s_{q,t}$, γ_q, and the positive-truncated Normal factors. Setting $Q = 1$, taking $\alpha_1 = \alpha$, $\boldsymbol{y}_{\mathrm{st}} = \boldsymbol{y}$, $\boldsymbol{Z}_{\mathrm{st}} = \boldsymbol{Z}$, $\boldsymbol{\alpha}_{\mathrm{st}} = \alpha \mathbf{1}_T$, $\boldsymbol{\beta}_{\mathrm{st}} = \boldsymbol{\beta}_1$, and dropping $\mathcal{B}_Q$ recovers the single-level posterior in (4.1). The exAL augmentation preserves the Gaussian slope-path update in (3.14)–(3.15), but the scale-asymmetry blocks (σ_q, γ_q) remain non-conjugate. We therefore use two computation routes for this fixed-design posterior: an MCMC sampler with separate exAL block updates and a mean-field variational Bayes approximation with Laplace–Delta updates for the non-conjugate blocks. In the algorithms below, GIG, positive-truncated Normal, Gaussian, and inverse-gamma entries denote closed-form full conditionals for MCMC and closed-form coordinate factors for VB. The same block structure reappears in the variational updates; only the final exAL scale-asymmetry block requires a non-conjugate treatment.

4.3 Markov Chain Monte Carlo

Sampler structure. The augmented likelihood and shrinkage hierarchy give a partially Gibbs sampler. Closed-form blocks are sampled from their named full conditionals; the complete-data targets, full conditionals, CAVI factors, and ELBO terms for AL and exAL likelihoods under ridge and RHS priors are given in Sections S5–S9 of the supplement. Under the Nishimura–Suchard hierarchy, the global-local scales have inverse-gamma updates. Each exAL block (σ_q, γ_q) remains non-conjugate and is updated with Metropolis–Hastings or slice moves on transformed coordinates. Algorithm 1 targets (4.2); with $Q = 1$, it uses the collapsed posterior in (4.1). Ridge and AL variants omit the corresponding shrinkage, $s_{q,t}$, and γ_q blocks. Diagnostics use trace plots, effective sample size summaries, crossing diagnostics for multi-quantile grids, and monitoring of the (σ_q, γ_q) updates, including acceptance rates when Metropolis–Hastings is used.

Algorithm 1 MCMC sampler for the Q–DESN posterior. For $Q = 1$, set $\boldsymbol{\beta}_{\text{base}} = \mathbf{0}_r$, place the RHS prior on $\Delta_1 = \boldsymbol{\beta}_1$, and omit baseline-slope blocks. AL and ridge variants omit the corresponding $s_{q,t}$, γ_q, and shrinkage blocks.

Require: Data $\{y_t, \tilde{\boldsymbol{x}}_t\}_{t=1}^{T}$, quantile grid $p_1 < \cdots < p_Q$, and priors
Require: MCMC iterations N and convergence-monitoring settings

```
Initialize α, β_st, likelihood parameters, latent variables, and RHS scales
if Q > 1 then
    Initialize β_base and its baseline RHS scales
end if
for i = 1 to N do
    Latent scales: for each q, t, draw v_{q,t} | · from the AL or exAL GIG full conditional
    exAL shifts: for exAL, draw s_{q,t} | · from its positive-truncated Normal full conditional; omit
        this step for AL
    Slope path: compute K_β and m_β from (3.14)–(3.15)
    Slope path: draw β_st | · ~ N(m_β, K_β^{-1})
    if Q > 1 then
        Baseline: draw β_base from its Gaussian conditional and draw its RHS scales from inverse-
            gamma full conditionals
    end if
    Intercepts: draw α from its Gaussian conditional, or update the ordered-gap block when
        exact intercept monotonicity is imposed
    RHS scales: draw adjacent-difference local, global, auxiliary, and slab scales from inverse-
        gamma full conditionals; for Q = 1, this is the ordinary single-level RHS block for β_1
    AL scale: for AL, draw each σ_q | · from its inverse-gamma full conditional and omit γ_q
    exAL scale–asymmetry: for exAL, update each (σ_q, γ_q) on transformed coordinates tar-
        geting its non-conjugate conditional kernel
    Monitor: store post-burn-in draws, crossing diagnostics, and convergence summaries, with
        thinning only if used
end for
```

4.4 Variational Approximation

VB–LD approximation. For lower-cost screening, warm starts, and companion validation panels, we also use a mean-field variational approximation to the same augmented target. The Gaussian, GIG, positive-truncated Normal, inverse-gamma, and RHS scale factors retain the closed-form coordinate structure implied by (4.2). The only extra non-conjugacy introduced by the exAL likelihood is the scale-asymmetry block (σ_q, γ_q). That block is approximated with the Laplace–Delta treatment for non-conjugate variational inference of Wang and Blei (2013), adapted to Q–DESN as detailed in the supplement. We therefore use the label VB–LD as a computational convention:

it identifies the variational approximation used for screening and initialization, while the MCMC sampler remains the reference posterior computation when simulation tables report MCMC results.

Equations (4.2) and (4.1) define the posterior distributions used for inference, and Algorithm 1 implements the corresponding MCMC updates. The supplement keeps the single-level full conditionals and the full quantile-vector posterior algebra explicit: Sections S5–S8 give the single-level posterior, MCMC, VB–LD, and ELBO updates, while Section S9 gives the joint quantile-vector posterior with regularized-horseshoe shrinkage over adjacent coefficient increments, stacked Gaussian blocks, and crossing diagnostics. For a single fitted quantile, Algorithm 1 uses $Q = 1$ and the collapsed RHS prior. For multiple fitted levels, the same sampler uses the baseline and adjacent-difference blocks. Generic VB–LD factors, ELBO monitoring, the corresponding coordinate-update algorithm, and the monotone rearrangement details are given in the supplement.

5 Multi-Step Quantile Forecasting and Scoring

Quantile paths and monotone rearrangement. For a forecast origin T, a quantile path is the sequence of fitted conditional quantiles over horizons $h = 1, \dots, H$. A single-level Q–DESN fit supplies this path for one probability level p_0. A grid of independent fits supplies one path for each fitted level $p_1 < \cdots < p_K$, but the resulting grid need not be monotone at every horizon. When a monotone finite-grid quantile curve is needed for display or scoring, the default rearrangement is the weighted isotonic projection

$$\boldsymbol{q}_t^* = \arg\min_{q_1 \le \cdots \le q_K} \sum_{k=1}^{K} w_k (\widehat{q}_{k,t} - q_k)^2, \qquad w_k > 0, \tag{5.1}$$

with equal weights unless a validation design specifies another choice in advance. The ordered grid is then linearly interpolated over $[p_1, p_K]$. Outside the fitted range, the article reports grid-level summaries unless an application explicitly states an endpoint convention. This rearrangement is a deterministic summary of fitted quantile levels. A full predictive distribution requires an additional interpolation, tail, and coupling construction.

5.1 Multi-Step Quantile Forecast Summaries

Forecast summaries are computed conditional on the fitted quantile-regression coefficients, the fixed reservoir construction, and the covariates available or specified at the forecast origin. Before forecasting, the reservoir state is obtained by running Eq. (2.1) through the observed record with observed response lags. After the forecast origin, future response lags are unknown; if they are needed for recursive multi-step forecasts, they must be generated within each Monte Carlo trajectory or supplied by a stated scenario.

For a single-level fit at probability p_0, let $\theta_{p_0}^{(m)} = (\boldsymbol{\beta}_{p_0}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)})$ denote a posterior or variational draw. Conditional on the draw-specific future design row, the fitted quantile location is

$$\mu_{p_0,T+h}^{(m)} = \boldsymbol{x}_{T+h}^{(m,p_0)\top} \boldsymbol{\beta}_{p_0}^{(m)}$$

for horizon h. These $\mu_{p_0,T+h}^{(m)}$ values are draws of the fitted p_0-level quantile location. If a scalar future response path is needed for recursive lags, the fitted AL or exAL working likelihood can be used to generate

$$y_{T+h}^{(m,p_0)} \sim \mathrm{exAL}_{p_0}\left(\mu_{p_0,T+h}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)}\right).$$

The resulting response draws are simulations from the fitted working likelihood. Summaries of $\mu_{p_0,T+h}^{(m)}$ describe posterior or variational uncertainty in the fitted quantile location. A closed-form marginal quantile of $Y_{T+h} \mid \mathcal{F}_T$ requires an explicitly stated marginalization or simulation rule.

For the joint quantile-vector regression, a draw contains level-specific coefficients $\{\alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q\}_{q=1}^{Q}$. The same future design vector then returns the raw fitted quantile vector

$$\mu_{q,T+h}^{(m)} = \alpha_q^{(m)} + \tilde{\boldsymbol{x}}_{T+h}^{(m)\top} \boldsymbol{\beta}_q^{(m)}, \qquad q = 1, \ldots, Q.$$

Let $\boldsymbol{q}_{T+h}^{(m)} = (\mu_{1,T+h}^{(m)}, \ldots, \mu_{Q,T+h}^{(m)})^\top$. This vector estimates the conditional quantile path over the fitted grid. A scalar response predictive distribution requires the separate propagation rule below.

If exact monotonicity is required before scoring or display, let $\mathcal{M}$ denote the stated monotone rearrangement, such as isotonic projection or rearrangement (Barlow and Brunk, 1972; Chernozhukov et al., 2010), and set

$$\boldsymbol{q}_{T+h}^{*,(m)} = \mathcal{M}\{\boldsymbol{q}_{T+h}^{(m)}\}.$$

The ordered grid defines a draw-specific quantile curve $Q_{T+h}^{*,(m)}(u)$ under the stated interpolation and endpoint convention. When recursive lags require a scalar future value, the implementation uses a specified quantile-curve sampling rule, described in the supplement:

$$u_{T+h}^{(m,r)} \sim \mathrm{Unif}(0,1), \qquad y_{T+h}^{(m,r)} = Q_{T+h}^{*,(m)}\left(u_{T+h}^{(m,r)}\right).$$

Thus the joint regression supplies the quantile curve used by the propagation rule. The scalar future response is generated from the reported quantile curve, using the stated interpolation and endpoint convention.

Single-level forecasts are scored by check loss at the fitted probability level. For a quantile grid, this article also uses aCRPS, defined as a finite-grid integrated check-loss score:

$$\mathrm{aCRPS}(y; \boldsymbol{q}) = 2\sum_{k=1}^{K} \omega_k \rho_{p_k}(y - q_k), \qquad \rho_p(e) = e\{p - \mathbf{1}(e < 0)\}.$$

The weights ω_k are the trapezoidal quadrature weights over the fitted probability grid. Thus aCRPS approximates the CRPS identity based on integrated quantile loss over the available grid. Full continuous-distribution CRPS requires a complete quantile function and the corresponding integral.

This distinction matters for the application tables below, where aCRPS compares finite sets of fitted quantile summaries.

6 Model Diagnostics and Reservoir Specification

Let

$$\mathcal{R} = \{D, \{n_d\}, \{\tilde{n}_d\}, m, T_0, \{\alpha_d\}, \{\rho_d\}, \{\pi_d^{(W)}\}, \{\pi_d^{(\mathrm{in})}\}, f, k, \mathrm{seed}\}$$

denote the depth, layer sizes, reduction sizes, lag memory, washout, leak rates, spectral radii, sparsity probabilities, activations, and random seed used to construct Eq. (2.1). Once $\mathcal{R}$ and the preprocessing convention are fixed, posterior inference is conditional on the resulting design matrix $\boldsymbol{X}$. Reservoir specification is therefore part of the empirical design: the Bayesian analysis estimates regression, likelihood, and shrinkage parameters conditional on the selected fixed feature map. Uncertainty over reservoir architectures would require an additional model layer outside the specification used here.

The ESN literature treats these choices as task-dependent design parameters. Reservoir size controls feature richness, leak rates and spectral radii jointly affect memory and stability, and input scaling determines how strongly the reservoir is driven into nonlinear regimes (Jaeger et al., 2007; Lukoševičius, 2012). For deep reservoirs, stacking layers can create hierarchical temporal representations, with additional depth justified by validation performance and stability diagnostics (Gallicchio et al., 2018). The condition $\rho_d < 1$ is a useful conservative design choice. For a driven nonlinear reservoir, empirical state diagnostics are still needed to assess stability (Yildiz et al., 2012). Several ESN modification papers identify failure modes that are visible before forecast scoring: output weights can reveal task-relevant reservoir units, state feedback changes the effective recurrent dynamics through an external feedback channel, redundant state trajectories can damage readout conditioning, and random reservoir realizations can induce substantial seed-to-seed variability (Fan et al., 2017; Ehlers et al., 2025; Saadat et al., 2025; Wu et al., 2018). In Q–DESN these results enter as diagnostics, sensitivity analyses, and possible ablations. After the feature map is fixed, posterior inference is for the quantile-regression parameters conditional on that feature map.

The empirical design fixes the training and validation windows, held-out test period, forecast origins, horizons, quantile levels, input transformations, and scoring rules before feature maps are compared. Numerical checks then verify input alignment, feature dimensions, washout handling, finite state values, empirical spectral radii, weakly varying or saturated reservoir states, and conditioning of the readout regression. State-matrix diagnostics include near-zero-variance columns, pairwise state correlations, singular-value or covariance spectra, effective rank, and a condition-number or eigenvalue-spread summary. These diagnostics distinguish a larger reservoir that adds nonredundant temporal features from one that mainly adds collinear columns to the fixed design matrix.

The relevant validation metric depends on the statistical question. In simulations with a known conditional quantile path, quantile-path RMSE, mean absolute error, bias, and correlation assess recovery of the true path. In both simulations and applications, forecast scores must use common forecast origins and horizons. For one fitted quantile level, check loss is the proper score for the reported quantile. For a quantile grid, interval score, empirical coverage, interval length, and aCRPS evaluate the finite set of fitted quantile summaries. Calibration of posterior credible bands under a misspecified working likelihood requires separate assessment.

The reservoir search used in the empirical sections varies depth, widths, lag memory, washout, leak rates, and spectral radii first; the supplement gives the full component-level checklist. Recurrent sparsity, input sparsity, input scaling, and activation choices are broadened when diagnostics indicate weak state variation, excessive saturation, or ill-conditioned design matrices. Leading specifications are compared over multiple reservoir seeds when feasible, and smaller diagnostic subsets compare VB with MCMC. The final specification is fixed before held-out evaluation. Reservoir ensembles, when used, are reported as sensitivity analyses across fixed feature maps.

Reproducible reporting gives the specification grid, defaults, state-quality summaries, validation origins, forecast horizons, scoring metrics, reservoir seeds, package and code versions, and final $\mathcal{R}$. Claims about a reservoir architecture are conditional on this validation design. Evidence from additional seeds, forecast origins, and scoring criteria is needed before generalizing beyond the reported comparison. State-feedback or principal-neuron reinforcement methods may be useful extensions; they change the feature-generation rule and require separate validation before being combined with the Q–DESN regression.

7 Simulation Validation Study

The main simulation evidence is organized around two complementary questions. The first asks whether a single-quantile Q–DESN regression can recover and forecast a known dynamic quantile path under a 500-observation benchmark following the family-based comparison strategy of Yan et al. (2025). The target quantile level is fixed in advance, the data-generating mechanism is known, and competing models are compared under matched error families, fit-window sizes, and quantile levels. The second asks whether a joint multi-quantile regression can recover several known conditional quantile paths under posterior simulation and maintain coherent raw quantile grids over a fixed set of levels. Companion VB screening and forecast-comparison tables are kept in the supplement rather than used as the main article evidence.

In contrast to the linear-regression setting of Yan et al. (2025), the oracle quantile paths are dynamic, and Q–DESN represents those paths through high-dimensional deterministic reservoir features. The design therefore separates two questions: how accurately a fitted model recovers the known conditional quantile path, and how well it forecasts held-out dynamic blocks. We write p for the single target level in the first study and τ for levels in the multi-quantile grid.

The main-text tables report MCMC summaries under a fixed validation design: simulated sources, rolling-origin metadata, fit and forecast windows, scoring rules, and model classes are specified before the reported comparisons are formed. The supplement gives VB companion panels and additional diagnostic summaries for the same validation setting. The single-quantile and multi-quantile studies use separate fixed designs, so their results are interpreted as complementary evidence rather than as one pooled experiment.

7.1 Single-Quantile Dynamic Data-Generating Processes

For each error family f and target quantile level $p \in \{0.05, 0.25, 0.50\}$, the synthetic series is generated as

$$y_{t,p,f} = \mu_{t,f} + \varepsilon_{t,p,f}, \qquad \varepsilon_{t,p,f} = e_{t,f} - H_f^{-1}(p),$$

where H_f is the distribution function of the raw error $e_{t,f}$. The shift by $H_f^{-1}(p)$ makes the p-quantile of $\varepsilon_{t,p,f}$ equal to zero. Hence

$$Q_p(y_{t,p,f} \mid \boldsymbol{\theta}_{t,f}) = \mu_{t,f}.$$

The oracle quantile path is generated by a six-dimensional dynamic linear state with local level, local slope, and two harmonic pairs:

$$\mu_{t,f} = \boldsymbol{F}^\top \boldsymbol{\theta}_{t,f}, \qquad \boldsymbol{F} = (1, 0, 1, 0, 1, 0)^\top,$$

with local linear trend and two sine–cosine harmonic pairs. The seasonal period is 90, the harmonics are 1 and 2, and the state innovation covariance $\boldsymbol{W}_\theta$ has diagonal square roots

$$\sqrt{\mathrm{diag}(\boldsymbol{W}_\theta)} = (0.005,\ 0.00002,\ 0.004,\ 0.004,\ 0.003,\ 0.003).$$

The initial covariance is $\boldsymbol{C}_0 = 0.01\boldsymbol{I}_6$, and each simulated trajectory is initialized deterministically at the family-specific state in Table 3. For a fixed family, the same latent path $\{\mu_{t,f}\}$ is used across the three values of p; only the centering constant $H_f^{-1}(p)$ changes.

Family	Raw error distribution H_f	Initial level	Initial slope	Seasonal settings
Gaussian	$\mathcal{N}(0, 10^2)$	40	0.012	harmonics $(24, 0.35)$, $(8, -0.80)$
Laplace	Laplace with scale 10	35	0.011	harmonics $(28, -0.15)$, $(10, 0.75)$
Gaussian mixture	$0.1\mathcal{N}(0, 0.5^2) + 0.9\mathcal{N}(1, 15^2)$	45	0.014	harmonics $(32, 0.85)$, $(12, -1.25)$

Table 3: Single-quantile dynamic data-generating mechanisms. Raw errors are shifted by $H_f^{-1}(p)$ so that the p-quantile of the observation error is zero; the same latent dynamic quantile path is then shared across target levels within each family. Seasonal entries report amplitude and phase for the two harmonic pairs in the initial state.

7.2 Single-Quantile Fit and Forecast Protocol

Each source trajectory contains $T_{\text{warm}} = 2000$ warmup values and $T_{\text{main}} = 10000$ main source values. All indices below refer to the main source scale after warmup removal. The fitting origin is source index 9000, and the held-out block is 9001:10000. The two fit-window sizes are $T_{\text{fit}} = 500$ and $T_{\text{fit}} = 5000$, corresponding to training windows 8501:9000 and 4001:9000. Forecast evaluation uses a rolling-origin state-update protocol over the held-out block. The primary protocol has maximum lead $H_{\max} = 30$ and origin stride 30. The reported tables use this pre-specified protocol for source identifiers, model specifications, and scoring metadata.

7.3 Single-Quantile Competing Methods

For each family and target quantile level in the single-quantile validation comparison, the reported Q–DESN rows vary the working likelihood and inference method while using regularized-horseshoe (RHS) coefficient shrinkage. Entries labeled Q–DESN AL–RHS use the AL working likelihood, whereas entries labeled Q–DESN exAL–RHS use the exAL working likelihood. Historical ridge fits are omitted from the main comparison because the reported validation focuses on Bayesian shrinkage fits under the train-window preprocessing rule. The reservoir specification, random seed, source identifiers, and software version are part of the supporting replication materials for every displayed metric. Dynamic quantile linear model (DQLM) and extended dynamic quantile linear model (exDQLM) baselines are fit to the same family, quantile level, fit-window, and forecast-origin protocol using exdqlm (version 1.0.0; Barata et al., 2026).

The primary comparison is a fixed-design benchmark against structured dynamic quantile linear models under linear dynamic mechanisms that favor those baselines. Repeated data-generating realizations, broader reservoir tuning, and non-Bayesian ESN regression baselines are left to the supplementary diagnostics and future sensitivity analyses.

7.4 Single-Quantile Criteria for Comparison

Let $\widehat{q}_t$ denote a posterior or variational summary of the fitted conditional p-quantile, such as the posterior median of the linear-predictor location. Because the oracle path μ_t is known, fit-window recovery can be evaluated using quantile mean absolute error, quantile RMSE, bias, correlation, and check loss at the target level p. Forecast accuracy is evaluated on held-out outcomes under common forecast origins and leads for every model. The reported tables show fit recovery and forecast scoring; runtime, row-completion, and source-identity diagnostics are available in the supporting replication materials.

For single-level quantile fits, the primary proper score is check loss at the fitted level. Quantile MAE and quantile RMSE compare the fitted or forecast quantile summary directly with the known oracle quantile path, while check loss scores realized observations relative to the reported quantile. These metrics therefore answer different questions and are reported separately. Runtime is reported as a computational diagnostic in the supplement.

Because the simulation uses one reproducible data-generating realization for each family and quantile level, the tables report deterministic criterion values for that source design rather than Monte Carlo means over repeated data sets. Any boldface or ranking in the displayed MCMC tables is interpreted within a fixed family, quantile level, fit-window, model family, and metric. The displayed Q–DESN values form a case-specific, metric-level summary: each entry is the best finite value observed for that model, family, quantile level, and metric. Consequently, entries within one model row may originate from different model specifications or replicate seeds. The contributing specification, diagnostic status, and source identifier for every displayed entry are recorded with the supporting replication materials.

7.5 Single-Quantile Fit and Forecast Results

Evaluation protocol. The verified protocol uses the 500-observation training window 8501:9000, the held-out block 9001:10000, rolling-origin forecasts with leads 1–30 and origin stride 30, and the three target levels $p = 0.05, 0.25, 0.50$. Tables 4–6 report the selected MCMC family-specific metric summaries. The entries are generated from lead-level rolling-origin interface rows: fit RMSE measures recovery of the known oracle quantile path on the training window, forecast MAE measures oracle-quantile forecast error averaged across scored rolling-origin lead-target pairs, and forecast check loss scores realized observations at the fitted quantile level. Runtime and implementation details are reported in the supplementary materials. Boldface marks the lowest displayed MCMC value within each quantile level and metric. Diagnostic status is disclosed alongside the scores: daggers mark rows with a warning in the supporting diagnostic summary, and double daggers mark rows with a failed diagnostic check.

All displayed Q–DESN and exQ–DESN values use feature scaling and other learned preprocessing quantities estimated from observations available through the fitting origin. The DQLM and exDQLM baselines use the same source trajectories, training window, forecast origins, leads, and scoring rules. The exAL–RHS MCMC rows use the v-augmented transition described in the supplement, which separates the scale update from the shape update and samples the shape on the logit transform of its bounded support.

Forecast criteria are computed from lead-level rolling-origin paths for every Q–DESN comparison. Thus fit RMSE, forecast MAE, and forecast check loss are all aligned with the same source trajectory, target level, training window, forecast origins, and scored leads. This convention gives a single reporting definition for Q–DESN, DQLM, and exDQLM within the 500-observation benchmark.

Figure 2 provides a cross-family view of the same comparisons. Its shading is normalized only within a fixed family, quantile level, and metric; the printed cell values retain the absolute scale. The colors therefore show relative departures from the best displayed value without combining RMSE, MAE, and check loss into a single criterion.

Displayed validation comparison. The displayed MCMC tables are the reported validation study for this fixed dynamic design. They are interpreted within the 500-observation setting, the three target levels, and the specified rolling-origin scoring protocol. Variational Bayes panels for the

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	3.036	3.829	1.103	3.811	2.209	3.328	2.590	**1.161**	**4.029**
exDQLM	**1.935**	16.23	4.313	2.344	6.634	4.132	2.808	1.637	4.109
Q–DESN AL–RHS†	2.779	8.410	1.221	2.176	**2.191**	**3.292**	1.523	2.104	4.073
Q–DESN exAL–RHS‡	2.149	**2.863**	**1.080**	**1.366**	2.415	3.317	**1.128**	1.971	4.059

Table 4: MCMC single-quantile fit-and-forecast comparison for the Gaussian simulation family. Q–DESN entries are reported within each model, target level, and criterion under the fixed validation design. When repeated-chain summaries are available, the displayed value uses the corresponding repeated-chain average. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostics; a double dagger marks a failed diagnostic check.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	**3.663**	9.791	1.968	2.850	3.520	4.548	1.774	1.337	5.082
exDQLM	5.677	22.44	5.154	1.710	6.811	5.106	1.774	2.013	5.206
Q–DESN AL–RHS†	5.420	3.893	1.873	2.256	1.434	4.379	1.353	0.7553	**5.008**
Q–DESN exAL–RHS†	6.409	**2.362**	**1.855**	**1.383**	**0.6696**	**4.343**	**1.075**	**0.6772**	5.008

Table 5: MCMC single-quantile fit-and-forecast comparison for the Laplace simulation family. Q–DESN entries are reported within each model, target level, and criterion under the fixed validation design. When repeated-chain summaries are available, the displayed value uses the corresponding repeated-chain average. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostics; a double dagger marks a failed diagnostic check.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	2.623	3.003	**1.505**	4.740	2.487	4.540	2.226	1.932	5.556
exDQLM	**2.554**	23.91	5.857	1.381	8.941	5.649	2.408	2.207	5.607
Q–DESN AL–RHS†	3.154	3.006	1.564	1.855	**0.9122**	**4.467**	1.271	**0.9074**	**5.441**
Q–DESN exAL–RHS‡	3.354	**2.913**	1.527	**1.380**	1.819	4.488	**0.9177**	2.562	5.610

Table 6: MCMC single-quantile fit-and-forecast comparison for the Gaussian mixture simulation family. Q–DESN entries are reported within each model, target level, and criterion under the fixed validation design. When repeated-chain summaries are available, the displayed value uses the corresponding repeated-chain average. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostics; a double dagger marks a failed diagnostic check.

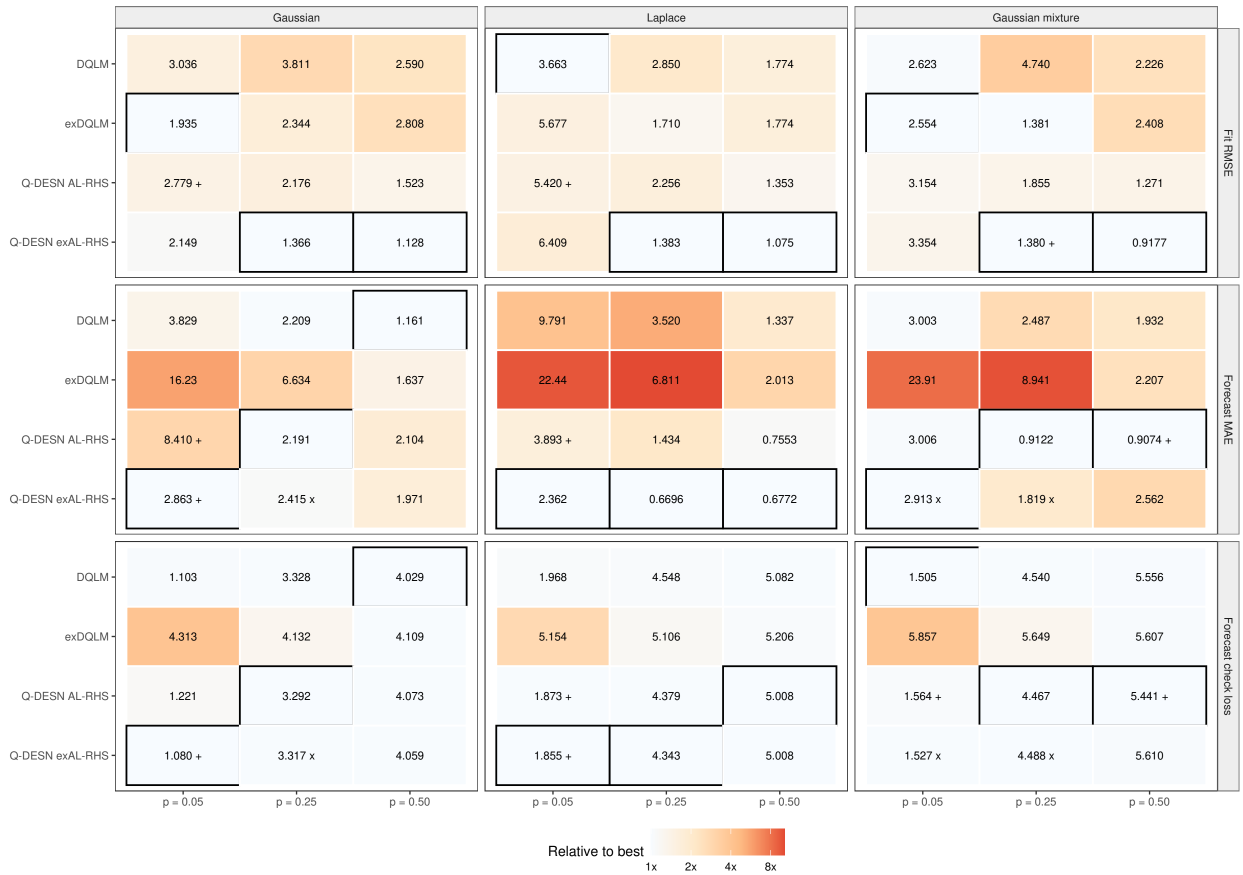

Figure 2: MCMC performance across the independent single-quantile simulation families. Columns identify the data-generating family and rows identify the criterion. Each cell prints the absolute criterion value, while shading gives its ratio to the lowest value among the four models for that family, quantile level, and criterion. Outlined cells are the within-comparison minima; darker cells indicate larger multiplicative departures. Plus signs and crosses mark values whose supporting diagnostics contain a warning or failed check, respectively. Entries are interpreted within each family, quantile level, and criterion; all criteria are lower-is-better. Forecast criteria use the held-out window of length 1000 with leads 1–30 and origin stride 30.

same source trajectories are reported in the supplement as computational companion evidence.

A five-chain sensitivity analysis for the Gaussian family at $p = 0.25$ is reported in the supplement. It assesses whether the selected Q–DESN behavior persists across independently seeded chains using a common posterior point-path summary.

7.6 Joint Multi-Quantile Validation with a Pre-Specified Feature Design

The second simulation study assesses whether a shared quantile-vector regression can recover several conditional quantile paths simultaneously under a feature design fixed before evaluation. Reservoir and prior specifications are chosen separately for each mechanism and model using VB or VB-LD. A replicated robustness experiment fits the 32 scenario–model specifications on 50 independently generated data-generating realizations per comparison, for 1,600 variational fits in total. The reported evidence is a balanced MCMC validation analysis of the same four regression–likelihood combinations: Joint QDESN RHS, Independent QDESN RHS, Joint exQDESN RHS, and Independent exQDESN RHS. Each MCMC run is initialized from its corresponding variational fit and uses the scenario-specific specification fixed before sampling. The comparison is restricted to posterior quantile-grid paths under the stated working likelihoods.

The exAL rows include eleven scale-collapsed fits whose quantile summaries satisfy the stability criterion and five stable exAL specifications selected under the same scenario-level validation design. Mixing qualifications are reported in the diagnostic summaries. Across the scale-collapsed exAL fits, fit-window MAE improves consistently, while forecast MAE improves in five of the sixteen exAL scenario–model comparisons.

The selected regression is fit across

$$\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}.$$

The balanced MCMC comparison grid uses eight synthetic data-generating mechanisms, a pre-specified feature design, and a 500-observation fit window. The three bridge mechanisms use Gaussian, Laplace, and Gaussian-mixture innovations. The five stress mechanisms introduce asymmetric tails, persistent heavy tails, Student-t errors, regime shifts, and nonlinear reservoir-friendly dynamics. The resulting grid contains 32 scenario–model MCMC rows. Fitted and forecast quantiles are compared with the known conditional quantile paths; realized observations are scored by check loss and finite-grid aCRPS; and hit-rate errors summarize marginal calibration. Reported scores use the pre-specified monotone rearrangement, while raw adjacent-level crossings before that rearrangement are retained as numerical coherence diagnostics.

Scenario-level results. Table 7 summarizes the balanced MCMC validation analysis at the scenario level. Entries report forecast MAE against the oracle quantile grid, with fit-window MAE and raw forecast-window crossing counts in parentheses; boldface marks the lowest within-scenario forecast MAE. Joint QDESN RHS is lowest in the Laplace, nonlinear, and normal mechanisms. Independent QDESN RHS is lowest in the asymmetric-tail, regime-shift, and Student-t location-scale

Scenario	Mechanism	Joint QDESN AL–RHS	Independent QDESN AL–RHS	Joint exQDESN exAL–RHS	Independent exQDESN exAL–RHS
Asymmetric Laplace Tail	Asymmetric Laplace; seasonal AR(1) location-scale	0.099 (0.084; 1)	**0.083 (0.088; 7)**	0.093 (0.081; 0)	0.089 (0.077; 0)
Gaussian-Mixture Bridge	Gaussian mixture; seasonal AR(1) location-scale	0.096 (0.097; 0)	0.099 (0.101; 0)	0.106 (0.096; 0)	**0.094 (0.095; 0)**
Laplace Bridge	Laplace; seasonal AR(1) location-scale	**0.085 (0.089; 0)**	0.094 (0.091; 7)	0.111 (0.113; 0)	0.106 (0.117; 0)
Nonlinear Reservoir Friendly	Gaussian mixture; nonlinear reservoir dynamics	**0.104 (0.109; 0)**	0.116 (0.112; 1)	0.157 (0.151; 0)	0.149 (0.143; 0)
Normal Bridge	Gaussian; seasonal AR(1) location-scale	**0.081 (0.082; 0)**	0.082 (0.082; 2)	0.115 (0.097; 0)	0.104 (0.101; 0)
Persistent Heavy Tail	Student-t; persistent seasonal AR(1) dynamics	0.131 (0.120; 0)	0.124 (0.118; 0)	0.130 (0.103; 0)	**0.106 (0.100; 0)**
Regime Shift	Student-t; regime-shift location-scale	0.153 (0.085; 0)	**0.096 (0.091; 6)**	0.185 (0.090; 0)	0.113 (0.087; 0)
Student-t Location-Scale	Student-t; seasonal AR(1) location-scale	0.080 (0.062; 0)	**0.071 (0.065; 1)**	0.116 (0.093; 0)	0.099 (0.092; 0)

Table 7: Scenario-level MCMC validation analysis for the joint multi-quantile validation study. Entries are forecast MAE with fit-window MAE and the number of raw forecast-window adjacent-level crossings in parentheses, written as forecast MAE (fit MAE; raw crossings). Boldface marks the lowest forecast MAE within each pre-specified eight-mechanism scenario. QDESN uses the AL working likelihood, exQDESN uses the exAL working likelihood, and RHS denotes the regularized horseshoe prior. All scores use the monotone quantile-grid reporting rule; raw crossings are retained only as pre-rearrangement diagnostics. exAL rows are included when finite-path and quantile-stability criteria are satisfied under the balanced validation design.

mechanisms, while Independent exQDESN RHS is lowest in the Gaussian-mixture and persistent-heavy-tail mechanisms. Joint exQDESN RHS has larger forecast MAE than at least one competing structure in each mechanism. These numerical lowest-scoring methods are descriptive because complete best-versus-runner-up Monte Carlo error is unavailable for all comparisons.

Coherence diagnostics and interpretation. Fit recovery gives a complementary result: Joint QDESN RHS has the lowest MAE in five of the eight fit windows, with Independent exQDESN RHS lowest in the three remaining cases. Quantile coherence also distinguishes the regression structures. Independent QDESN RHS accounts for 24 of the 25 raw forecast-window crossings; Joint QDESN RHS accounts for one, and both exAL rows account for none. All 32 rows have zero crossings after the monotone rearrangement. The replicated VB experiment further gives lower median paired forecast MAE to AL than exAL for both regression structures in every mechanism, although the frequency and size of the differences vary by scenario. Together, these results identify Joint QDESN RHS as the principal joint-regression reference in this design. The exAL shape parameter is useful in some comparisons, but its forecast quantile-path accuracy is heterogeneous across mechanisms. The supplement reports the computational protocol, diagnostic criteria, metric-specific lowest-scoring methods, and replicated AL–exAL contrasts.

8 Application: GloFAS Retrospective Streamflow Quantile Forecasting

Retrospective case study. The empirical application considers a medium-range retrospective streamflow quantile-forecasting case using the Global Flood Awareness System (GloFAS) and a reference gauge record. GloFAS is a global hydrological forecast and monitoring system jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts (ECMWF) and operated within the Copernicus Emergency Management Service (Alfieri et al., 2013; Copernicus Emergency Management Service, 2026). Its operational forecast and reforecast products provide ensemble river-discharge information for flood early warning and forecast verification (Harrigan et al., 2023). Previous calibration work has focused on hydrological model parameters using daily streamflow observations (Hirpa et al., 2018); here Q–DESN is used as a statistical quantile-regression correction for one forecast system and one specified forecast origin. The case is retrospective: the selected specification uses a blended future-covariate information set and is evaluated at a single origin.

8.1 Data and Forecast-Origin Design

Let T denote a forecast origin and let H be the issued forecast horizon. The historical reference record is y_t, the transformed USGS streamflow through time T. The retrospective GloFAS product over the same period is g_t^{ret}, and $g_{T+h,m}^{\text{ens}}$ denotes GloFAS ensemble member m issued at origin T for horizon $h = 1, \ldots, H$. The response values $y_{T+1}, \ldots, y_{T+H}$ are held out during prediction and used later for validation.

Retrospective information set. The reported analysis uses the forecast origin 2022-12-25 and scores seven fitted quantile levels over 28 issued target dates. The application is a single-origin retrospective case. Its response values are held out for scoring, and the selected covariate policy includes realized-history and blended Global Ensemble Forecast System (GEFS) forecast information with realized-future weather contributions for precipitation and soil-moisture covariates. This restriction is intentional: it keeps the data extraction, forecast-origin alignment, monotone quantile summaries, and model-selection summary reproducible before broader comparisons are reported.

8.2 Application Model

Persistence-anchored discrepancy. For $t \leq T$, the application uses two deterministic Q–DESN feature maps. The reference block $\boldsymbol{x}_t^{\text{ref}}$ is driven by the transformed reference history and the selected covariate policy. The discrepancy block $\boldsymbol{x}_t^{\text{disc}}$ is driven by the retrospective discrepancy $g_t^{\text{ret}} - y_t$, using the same documented preprocessing, lag, reservoir, reducer, and washout conventions. For $t > T$, the selected retrospective specification uses the persistence-anchored innovation rule: the forecast-window discrepancy feature is anchored at the last historical retrospective discrepancy and the discrepancy regression models an adjustment around that persistence baseline. The empirical issued-ensemble quantile is retained as a raw GloFAS comparator and as part of the forecast-system evidence table; the selected forecast-window discrepancy-reservoir input is the persistence-anchored discrepancy state.

For a fixed quantile level p_0, the reference-process quantile is represented by one Q–DESN regression and the GloFAS discrepancy by a second regression,

$$q^{\mathrm{ref}}_{p_0,t} = \boldsymbol{x}^{\mathrm{ref}\top}_t \boldsymbol{\beta}_{p_0}, \qquad d^{\mathrm{glo}}_{p_0,t} = \boldsymbol{x}^{\mathrm{disc}\top}_t \boldsymbol{\alpha}_{p_0}, \qquad q^{\mathrm{glo}}_{p_0,t} = q^{\mathrm{ref}}_{p_0,t} + d^{\mathrm{glo}}_{p_0,t}.$$

The path $q^{\mathrm{ref}}_{p_0,t}$ is the reference-process quantile of interest, while $d^{\mathrm{glo}}_{p_0,t}$ is the quantile-specific discrepancy between GloFAS and the reference process. The retrospective and ensemble GloFAS products inform the forecast-system quantile path $q^{\mathrm{glo}}_{p_0,t}$; the scored reference quantity remains $q^{\mathrm{ref}}_{p_0,T+h}$.

At forecast origin T, the issued GloFAS ensemble supplies river-discharge alternatives from the forecast system at each covered target date. The reported application summary is a VB quantile-regression summary of the reference process and the persistence-anchored discrepancy adjustment. For variational draw or summary index $s = 1, \dots, S$, the construction records the reference quantile and the resulting forecast-system quantile:

$$q^{\mathrm{ref},(s)}_{p_0,T+h} = \boldsymbol{x}^{\mathrm{ref},(s)\top}_{T+h} \boldsymbol{\beta}^{(s)}_{p_0}, \qquad q^{\mathrm{glo},(s)}_{p_0,T+h} = q^{\mathrm{ref},(s)}_{p_0,T+h} + d^{\mathrm{ret}}_T + \boldsymbol{x}^{\mathrm{disc},(s)\top}_{T+h} \boldsymbol{\alpha}^{(s)}_{p_0}.$$

Here d^{ret}_T is the last historical retrospective discrepancy on the transformed scale. Posterior means, medians, intervals, and monotone multi-quantile summaries are computed after this quantile construction.

The working-likelihood specification is

$$\begin{aligned}
y_t \mid q^{\mathrm{ref}}_{p_0,t}, \sigma_{\mathrm{ref}}, \gamma_{\mathrm{ref}} &\sim \mathrm{exAL}_{p_0}(q^{\mathrm{ref}}_{p_0,t}, \sigma_{\mathrm{ref}}, \gamma_{\mathrm{ref}}),\\
g^{\mathrm{ret}}_t \mid q^{\mathrm{glo}}_{p_0,t}, \sigma_{\mathrm{glo}}, \gamma_{\mathrm{glo}} &\sim \mathrm{exAL}_{p_0}(q^{\mathrm{glo}}_{p_0,t}, \sigma_{\mathrm{glo}}, \gamma_{\mathrm{glo}}),\\
g^{\mathrm{ens}}_{T+h,m} \mid q^{\mathrm{glo}}_{p_0,T+h}, \sigma_{\mathrm{glo}}, \gamma_{\mathrm{glo}} &\sim \mathrm{exAL}_{p_0}(q^{\mathrm{glo}}_{p_0,T+h}, \sigma_{\mathrm{glo}}, \gamma_{\mathrm{glo}}), \quad h = 1, \dots, H, \quad m = 1, \dots, M.
\end{aligned}$$

This display gives the general exAL form. The AL specialization used by the selected AL analysis omits s-latent variables and γ blocks, and replaces the exAL constants by the corresponding AL constants at the fitted level. The issued ensemble members enter through a working likelihood that treats them as conditionally independent and exchangeable given the forecast-system quantile path and source-specific likelihood parameters. ECMWF ensemble forecasts use perturbed initial states and model perturbations to produce distinct forecast alternatives (European Centre for Medium-Range Weather Forecasts, 2026); the conditional-independence factorization is therefore a modeling approximation for the Q–DESN regression layer. In exAL variants, the default application model shares σ_{glo} and γ_{glo} between retrospective and issued-ensemble GloFAS rows; under the AL specialization, only the corresponding σ_{glo} block is shared.

For this retrospective analysis, the seven fitted quantile levels are 0.05, 0.15, 0.35, 0.50, 0.65, 0.80, and 0.95. They are fitted independently with the AL special case and the VB approximation. The streamflow application is intended as a fast retrospective quantile-regression analysis, so the reported run uses Q–DESN under AL with independent fitting and monotone rearrangement. The variational quantile summaries are passed through the monotone rearrangement to form the displayed intervals.

No additional spread calibration is applied before display; the reservoir features, likelihood, priors, and fitted regression coefficients remain those of the selected fit.

The selected fit uses separate DESN feature maps for the reference quantile path and the GloFAS discrepancy path. Both maps have depth $D = 1$, layer widths (300), reducer widths (*none*), memory 360, washout 500, spectral radii (0.95), recurrent sparsity (0.03), and input inclusion probabilities (1). The reference map uses leak rates (0.1) and global/bias input scales 0.18/0.18, and reservoir seed 20260512; the discrepancy map uses leak rates (0.1), global/bias input scales 0.18/0.18, and reservoir seed 20261521. Forecast-window discrepancy states use the persistence-anchored innovation transition.

The two coefficient blocks use separate regularized-horseshoe global scales, $\tau_{0,\text{ref}} = 0.1$ and $\tau_{0,\text{disc}} = 0.001$, respectively. A configuration summary documents the data version, preprocessing choices, selected files, and fitted specification for this retrospective analysis.

8.3 Retrospective Application Results

The selected fit was evaluated over 12,495 observed dates through the forecast origin. After monotone rearrangement, its observational-window aCRPS was 0.0445, and empirical 90% interval coverage was 0.906. This evidence governs the specification choice; the forecast window remains a held-out assessment. The retrospective diagnostic evaluation also identified occasional extreme fitted values in the $p_0 = 0.95$ path, so the upper-tail component remains subject to targeted sensitivity analysis.
Forecast-window scores. Table 8 reports the forecast scores for the reported retrospective analysis. Relative to raw GloFAS, Q–DESN reduces the mean check loss from 0.7639 to 0.5654, the interval score from 25.2028 to 10.8032, and aCRPS from 1.4424 to 1.1319. Mean empirical interval coverage increases from 0.000 to 0.357, but the Q–DESN value remains far below the nominal 90% level. These summaries describe one specified forecast origin under a blended retrospective information set. Multi-origin operational validation and calibrated predictive intervals require broader experiments.

Table 8: Forecast scoring for the GloFAS retrospective analysis. Scores are computed on the transformed streamflow scale for the forecast origin 2022-12-25. Lower check loss, interval score, and aCRPS are better; aCRPS is the finite-grid trapezoidal integrated check-loss score over the seven fitted levels. Mean coverage reports empirical coverage for the nominal 90% reported interval and is interpreted descriptively. The final column gives the relative reduction in mean check loss.

Model	Horizons	Check	Interval	aCRPS	Coverage	Check reduction
Q–DESN correction	28	0.5654	10.8032	1.1319	0.357	26.0%
Raw GloFAS	28	0.7639	25.2028	1.4424	0.000	Reference

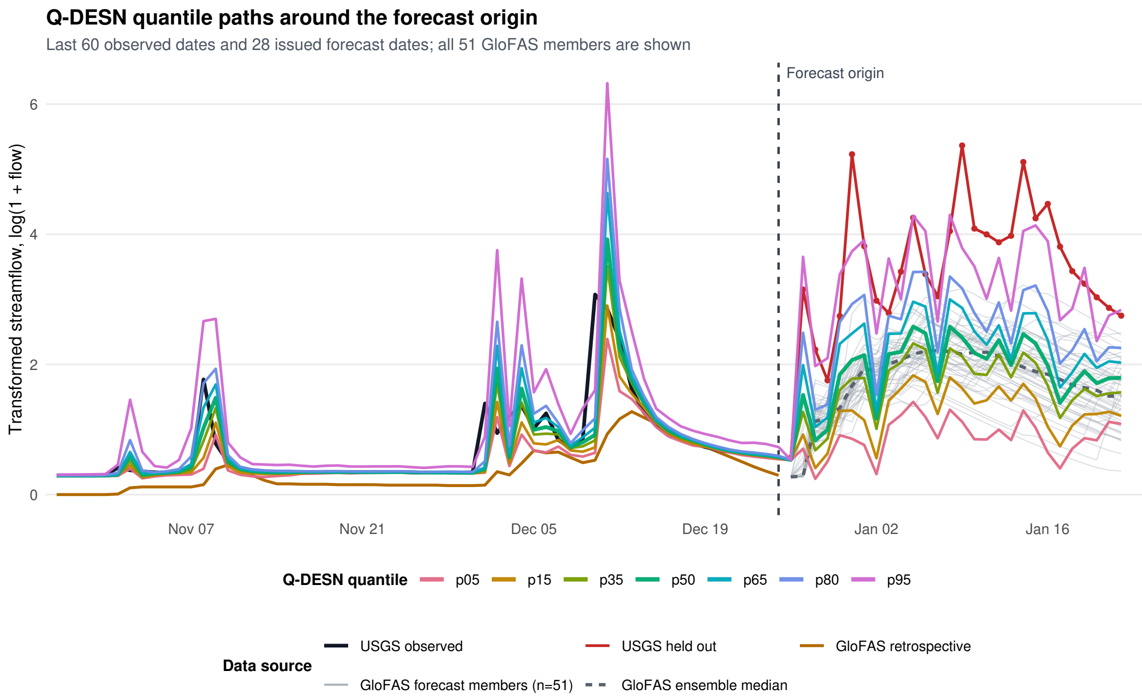

Figure 3: Discrepancy-corrected Q–DESN quantile paths for the GloFAS retrospective analysis. The figure shows the final 60 observed dates through the forecast origin (vertical dashed line) and the following 28-day issued forecast window. Black denotes observed USGS streamflow before the origin, red denotes the held-out USGS measurements, and orange denotes retrospective GloFAS. Thin gray paths are all 51 issued GloFAS ensemble members; their median is the dashed gray path.

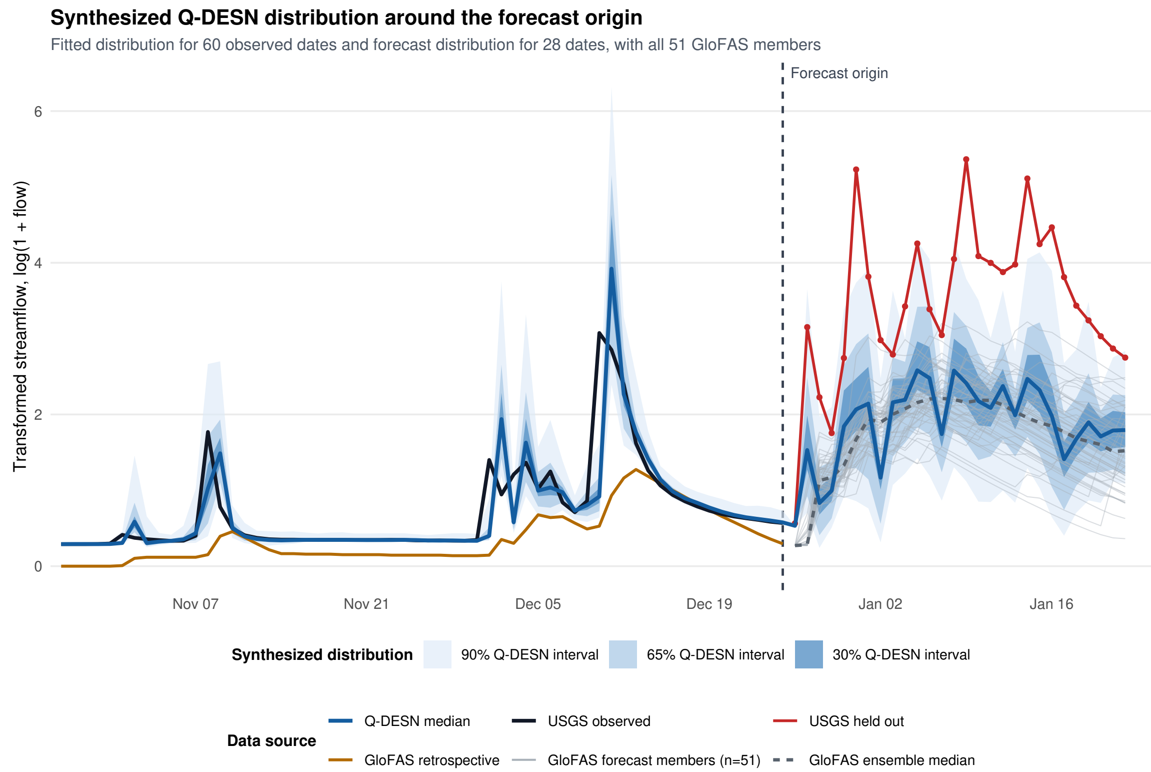

Figure 4: Synthesized Q–DESN quantile bands for the GloFAS retrospective analysis over the final 60 observed dates and the following 28-day forecast window. The nested blue bands are the 90%, 65%, and 30% quantile intervals, and the blue path is the median. The vertical dashed line marks the forecast origin. Black and red paths denote observed and held-out USGS streamflow, respectively; orange denotes retrospective GloFAS, and thin gray paths show all 51 issued GloFAS ensemble members, with their median dashed. Forecast scores use the held-out measurements in Table 8.

Reproducibility checks confirm the data extraction, selected files, score table, and VB convergence summary used for this retrospective analysis. The VB fits required a median of 103 and a maximum of 356 iterations.

9 Application: PriceFM Retrospective Electricity-Price Comparison

Benchmark quantity. As a second empirical case, we consider a retrospective comparison of probabilistic day-ahead electricity-price forecasts using version 4 of the PriceFM benchmark (Yu et al., 2026). The current benchmark release spans 1 January 2022 through 1 January 2026. The scored quantity is original-unit quantile loss on the PriceFM paper quantile grid under fold-aligned test timestamps. The released data cover European bidding regions at 15-minute resolution and include day-ahead price, load, solar, and wind forecasts. PriceFM itself is a graph-informed foundation model: it uses region-level exogenous information and a sparse graph mask derived from the European transmission topology. The reported comparison is a 114 region–fold comparison against fold-aligned PriceFM predictions. It evaluates the aligned benchmark panel used in this article.

9.1 Comparison Design

The reported benchmark contains 114 region/fold evaluations, spanning 38 European bidding regions and 3 rolling folds. Each evaluation is scored on the same fold-aligned test timestamps used by the PriceFM predictions. The scoring metric is original-unit average quantile loss (AQL) on the PriceFM paper quantile grid $0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90$. Throughout this section, Δ AQL denotes Q–DESN AQL minus PriceFM AQL. Negative values indicate lower Q–DESN AQL; positive values indicate lower PriceFM AQL.

Retrospective covariates. The benchmark panel is assembled from selected Q–DESN analyses and precomputed paper-quantile score files. The assembly uses the existing PriceFM predictions. Across the benchmark, 58 selected region/fold evaluations use own-region predictors and 56 selected region/fold evaluations add PriceFM neighborhood summaries. Both input policies use retrospectively observed own-region load, solar, and wind lead covariates in the comparison; evaluations with neighborhood summaries add retrospectively observed neighboring-region lead summaries. The score comparison is therefore retrospective and fold-aligned. The benchmark is interpreted as an application of the selected Q–DESN quantile-regression analyses. A PriceFM study of the joint quantile-vector RHS prior would require the same fold-aligned design with that joint regression fitted and selected directly.

PriceFM reports two model-comparison protocols. Its Table II is a full-shot evaluation, whereas its Table III evaluates leave-one-region-out zero-shot generalization. Because the Q–DESN specifications here are fitted and selected with observations from each target region, the full-shot table is the relevant paper-level analogue. Ranking for the present region-specific fits is therefore based on the full-shot comparison.

9.2 Full-Surface Results

Aggregate comparison. Across the aligned benchmark, Q–DESN has lower AQL in 66 of 114 region/fold evaluations (57.9%), 12 evaluations are near-equivalent under the pre-specified near-equivalence threshold, and PriceFM has lower AQL in 36 evaluations. The mean Q–DESN AQL is 6.826, compared with mean PriceFM AQL 7.039, giving mean Δ AQL -0.213 and median Δ AQL -0.083. The aggregate difference is therefore modest and heterogeneous. The fold-level supplementary summary shows that the mean Δ AQL is most negative in fold 2, modestly negative in fold 1, and essentially neutral in fold 3. On the same 114-comparison panel, the region–fold macro mean AQCR is 2.64% for Q–DESN and 0.00% for the released PriceFM checkpoint. Mean MAE is 16.728 versus 17.274, and mean RMSE is 25.453 versus 26.336, respectively. Thus Q–DESN reduces aligned AQL, MAE, and RMSE by approximately 3.0%, 3.2%, and 3.4%, while retaining a nonzero quantile-crossing rate.

Table 9: PriceFM-aligned retrospective model comparison. The first panel reports direct region–fold macro means from the common 38-region, three-fold comparison; boldface marks the better of those two directly aligned rows. The second panel reproduces the full-shot values in Table II of PriceFM version 4 (Yu et al., 2026). The paper-reported panel uses a distinct fitted-model and aggregation pipeline and is therefore contextual for the aligned comparison panel. AQL, MAE, and RMSE are in EUR/MWh; AQCR is in percent. Rank is reproduced only for the paper-reported panel; dashes denote quantities without a cross-panel rank. Lower is better. The comparison uses retrospectively observed own-region covariate leads in all selected rows and additional retrospectively observed neighboring-region leads when neighborhood summaries are included.

	Probabilistic		Pointwise		
Model	AQL	AQCR (%)	MAE	RMSE	Rank
Direct fold-aligned comparison (this article; 114 region/fold evaluations)					
Selected Q–DESN	**6.83**	2.64	**16.73**	**25.45**	–
Released PriceFM checkpoint	7.04	**0.00**	17.27	26.34	–
PriceFM v4 Table II (paper-reported full-shot evaluation)					
Naïve^1	15.29	0.00	22.06	34.68	11
Naïve^2	15.35	0.00	23.31	34.31	13
Naïve^3	15.46	0.00	22.64	32.61	12
FEDFormer	8.22	15.33	20.15	31.75	8
PatchTST	8.06	18.21	20.20	31.59	8
iTransformer	8.24	13.96	21.03	32.11	9
TimesNet	7.98	13.42	19.48	30.94	7
TimeXer	8.30	14.77	21.94	31.88	10
GraphConv	6.61	6.88	16.81	25.97	4
GraphAttn	7.13	10.33	17.00	26.11	6
GraphSAGE	6.78	6.01	17.56	26.03	5
GraphDiffusion	6.69	5.72	16.44	25.93	2
GraphARMA	6.72	6.03	16.56	25.84	3
PriceFM	**5.80**	**0.00**	**14.28**	**22.39**	**1**

The directly aligned checkpoint is the appropriate distribution-level comparator because its predictions are aligned row by row with the Q–DESN predictions. Its AQL differs from the PriceFM paper's optimized full-shot AQL, so the two PriceFM rows are kept separate. Numerically, the aligned Q–DESN AQL lies within the range reported for the graph baselines in PriceFM Table II, but differences in fitted objects and aggregation preclude a cross-panel rank claim. The fold-level win, tie, and loss summary is reported as a supplementary diagnostic.

The input set also matters. Evaluations with neighborhood summaries have the most negative average Δ AQL, whereas own-region-only evaluations are closer to neutral. Because input sets and selected model specifications vary across region/fold evaluations, the input-set comparison is descriptive. It summarizes where the selected Q–DESN analyses are strongest on the aligned benchmark. Input-set and horizon-block breakdowns are reported in the supplement.

9.3 Horizon Diagnostics and Remaining Gaps

Horizon-block diagnostics are available for 72 of the 114 region/fold evaluations. These diagnostics preserve the same convention as the aggregate comparison: negative Δ values indicate lower Q–DESN AQL. They should be read as a partial diagnostic summary for the subset with retained horizon-block scores.

Remaining gaps. The resulting evidence supports a conservative interpretation. In this retrospective comparison, Q–DESN is a competitive fixed-reservoir quantile-regression baseline against fold-aligned PriceFM predictions on the aligned benchmark panel, with lower mean and median AQL and lower region–fold macro MAE and RMSE in aggregate. This descriptive advantage is relative to the released checkpoint comparison. The separately reported optimized full-shot PriceFM result is contextual. A paired uncertainty interval is left for a future benchmark analysis. The advantage is heterogeneous: fold 3 is nearly tied on average, and PriceFM has lower AQL for a nontrivial subset of region/fold evaluations. These PriceFM-favored cases are retained as negative diagnostic evidence for the present Q–DESN design. A new PriceFM comparison would require a stated change to the information set, model class, selection rule, or joint quantile-regression design. A fully joint PriceFM run would be needed before making application-level claims about the joint multi-quantile regression model.

10 Discussion

Q–DESN treats the fixed DESN reservoir as a nonlinear feature map and places Bayesian quantile inference on the regression coefficients. This separation is useful computationally: posterior simulation and VB update only the coefficients, likelihood, and shrinkage parameters, while the recurrent weights, reducers, scaling constants, and washout convention remain fixed. The empirical results therefore evaluate a conditional Bayesian quantile-regression analysis under a specified reservoir construction.

The likelihood is also deliberately limited in interpretation. The AL and exAL families are used as quantile-targeting working likelihoods. They provide posterior distributions and conditionally tractable augmented updates. The simulation and application sections report fit recovery, forecast scoring, empirical predictive coverage, runtime, convergence, and reproducibility diagnostics to assess the model-based summaries under possible misspecification. Sandwich adjustments and other general-Bayes calibration checks remain important sensitivity directions, especially for composite multi-quantile likelihoods.

The multi-quantile construction makes the same distinction. Independent single-level Q–DESN fits followed by isotonic regression or rearrangement produce monotone quantile curves from level-wise regressions. The joint quantile-vector extension shares information across quantile levels through RHS shrinkage on adjacent coefficient differences, giving a soft non-crossing mechanism. Exact monotonicity for downstream use requires an ordered-intercept parameterization, projection, rearrangement, or another explicitly specified monotone rearrangement.

The validation evidence supports correspondingly limited claims. The single-quantile dynamic simulations test recovery and held-out forecasting under fixed data-generating mechanisms. The joint multi-quantile synthetic study checks coherence under known conditional quantile paths. The GloFAS application is a one-origin retrospective streamflow quantile-forecasting case using computationally light independent Q–DESN AL–VB fits, blended future weather covariates, monotone reporting, and an undercovered nominal 90% interval. The PriceFM comparison is a retrospective fold-aligned comparison against stored benchmark predictions with retrospectively observed lead covariates. These studies support the reported analyses under their own evaluation designs. Broader rankings of Q–DESN against dynamic quantile or foundation-model alternatives require additional multi-design comparisons.

Several extensions are natural. On the statistical side, future work should study calibrated composite-likelihood scaling, hard or probabilistic non-crossing constraints, richer covariance structures for quantile differences, and sensitivity to reservoir scaling on bounded feature domains. On the computational side, the joint model calls for sparse precision solvers, selected covariance extraction for VB, and diagnostics comparing VB with MCMC on small joint problems. On the application side, deployable multi-origin GloFAS studies with origin-available covariates, paired PriceFM uncertainty analyses, and explicitly joint PriceFM runs are needed before the joint quantile-vector prior can be interpreted in an application-level benchmark.

Within these boundaries, the article establishes Q–DESN as a reproducible fixed-feature Bayesian quantile-regression analysis whose inference, monotone rearrangement, diagnostics, and empirical validation are tied to explicit evaluation designs.

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